

AI-Driven Employee Retention Platform

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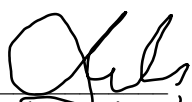
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AI-Driven Employee Retention Platform

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ABSTRAK

Burnout and lack of recognition are two primary drivers of employee turnover, representing a significant operational issue for Small to Medium Enterprises (SMEs). They should be detected early to prevent expenses of high financial and productivity. Conventional forms of management that are based on manual supervision and annual review are likely to lead to subjective evaluation and a long-detection process. The proposed project is the AI-Driven Employee Retention Platform, which is a web-based application that helps managers and HR professionals with employee retention based on rule-based logic and machine learning applications.

There is a rule-based engine within the system that is capable of predicting the risk of burnout precisely on the basis of the key performance measures, such as the task load and the compliance with the due-date. This performance data can be tracked through the interface of the system and analyzed to generate proactive burnout warnings and recognition recommendations to under-appreciated high-performers. This platform promotes the efficiency of management by identifying risks automatically, providing credible access to data on employee well-being, and eventually making a predictive model of overall turnover risk.

The project entails the creation of a full web-based platform as well as a machine learning model (trained on synthetic data), which was then evaluated through extensive testing of its predictive performance and user experience. This research paper validates that the system improves the speed of detection and the accuracy of diagnosis of the prevalent retention risks, including burnout and lack of recognition, as well as enables the successful and data-driven management of SMEs.

ABSTRACT

Burnout and lack of recognition are two primary drivers of employee turnover, representing a significant operational issue for Small to Medium Enterprises (SMEs). They should be detected early to prevent expenses of high financial and productivity. Conventional forms of management that are based on manual supervision and annual review are likely to lead to subjective evaluation and a long-detection process. The proposed project is the AI-Driven Employee Retention Platform, which is a web-based application that helps managers and HR professionals with employee retention based on rule-based logic and machine learning applications.

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TABLE OF CONTENT

ACKNOWLEDGEMENTS	ii
ABSTRAK	iii
ABSTRACT	iv
TABLE OF CONTENT	v
LIST OF TABLES	viii
LIST OF FIGURES	ix
LIST OF SYMBOLS	xi
LIST OF ABBREVIATIONS	xii
LIST OF APPENDICES	xiii
 CHAPTER 1	 14
1.1 Introduction	14
1.2 Problem Statement	15
1.3 Objectives	16
1.4 Scope	17
 CHAPTER 2	 19
2.1 Introduction	19
2.2 Existing Systems	20
2.2.1 Odoo	20
2.2.2 OrangeHRM	22
2.2.3 ERPNext	24
2.3 Comparison of Existing System	26
2.3.1 Relevance of Comparison with Project Title	28
2.4 Summary	29

CHAPTER 3	30
3.1 Introduction	30
3.2 Project Management Framework / Methodology	31
3.2.1 Agile Software Development Life Cycle (Agile SDLC).	31
3.3 Project Requirement	38
3.3.1 Functional Requirements	38
3.3.2 Non-Functional Requirements	39
3.3.3 Constraints and Limitations	40
3.4 Propose Design	41
3.4.1 Context Diagram	41
3.4.2 Use Case Diagram	43
3.4.3 Use Case Description	44
3.5 Data Design	51
3.5.1 ERD	51
3.5.2 Database Dictionary	52
3.6 PROOF OF INITIAL WORK	57
3.7 Testing/Validation Plan	67
3.8 Potential Use of Proposed Solution	71
 CHAPTER 4	 72
4.1 INTRODUCTION	72
4.2 Implementation & Development	72
4.2.1 Development Process & Tools	73
4.3 System Components & Functions	74
4.3.1 Login	74
4.3.2 Admin Dashboard	75

4.3.3	User Management page	76
4.3.4	Turnover/Need Help Alerts Page	77
4.3.5	Recognition Alerts Page	77
4.3.6	Leave Management Page	78
4.3.7	Dashboard Page	79
4.3.8	Project Pages	80
4.3.9	Need Help Alerts Page	82
4.3.10	Team Leave Request Page	84
4.3.11	Employee Dashboard Page	85
4.3.12	My Tasks Page	86
4.3.13	My Leave Page	88
4.4	Testing	89
CHAPTER 5		91
REFERENCES		92

LIST OF TABLES

Table 2.1 Comparison of Existing System	26
Table 3.1 Manage Authentication	44
Table 3.2 Manage Users	45
Table 3.3 Manage Projects	46
Table 3.4 Manage Tasks	47
Table 3.5 Manage Task Ratings	48
Table 3.6 Manage Leave Requests	49
Table 3.7 Manage Alerts	50
Table 3.8 USER TABLE	52
Table 3.9 PROJECT TABLE	53
Table 3.10 TASK TABLE	53
Table 3.11 ALERT TABLE	55
Table 3.12 LEAVE_REQUEST TABLE	56
Table 3.13 (UAT) Test Cases	69
Table 4.1 Development Process & Tools	73
Table 4.2 Functional Testing	89

LIST OF FIGURES

Figure 2.1 Odoo main system plugins	21
Figure 2.2 Odoo install plugins page	21
Figure 2.3 Odoo AI plugins	21
Figure 2.4 OrangeHRM	23
Figure 2.5 ERPNext Layout	24
Figure 2.6 ERPNext account function	25
Figure 3.1 Agile Software Development Life Cycle	31
Figure 3.2 Design Structure	33
Figure 3.3 Context Diagram	41
Figure 3.4 Use Case Diagram	43
Figure 3.5 ERD	51
Figure 3.6 Login page	57
Figure 3.7 Admin/HR Dashboard Page	58
Figure 3.8 User management Page	58
Figure 3.9 Turnover Alert Page	59
Figure 3.10 Burnout Alert Page	59
Figure 3.11 Recognition Alert Page	60
Figure 3.12 leave page	60
Figure 3.13 Manager Dashboard Page	61
Figure 3.14 Tasks page	61
Figure 3.15 Add new project pop up	62
Figure 3.16 Burnout alert page	62
Figure 3.17 Recognition page	63
Figure 3.18 Leave page	63
Figure 3.19 Employee dashboard	64
Figure 3.20 My tasks available projects page	64
Figure 3.21 My tasks inside project page	65
Figure 3.22 My tasks change status page	65
Figure 3.23 My tasks change status successful message	66
Figure 3.24 My leave page	66
Figure 4.1 Login Page	74
Figure 4.2 Admin Dashboard Page	75
Figure 4.3 Users Management Page	76

Figure 4.4 Turnover/Need Help Alerts Page	77
Figure 4.5 Recognition Alerts Page	77
Figure 4.6 Leave Management Page	78
Figure 4.7 Dashboard Page	79
Figure 4.8 project page	80
Figure 4.9 Tasks Page	81
Figure 4.10 Need Help Alerts Page	82
Figure 4.11 Recognition Page	83
Figure 4.12 Team Leave Request page	84
Figure 4.13 Employee Dashboard Page	85
Figure 4.14 My Tasks Page	86
Figure 4.15 Inside My Tasks page	87
Figure 4.16 My Leave Page	88

LIST OF SYMBOLS

LIST OF ABBREVIATIONS

LIST OF APPENDICES

CHAPTER 1

1.1 Introduction

This project develops the AI-Driven Employee Retention Platform, a web-based decision support system designed for Small and Medium Enterprises (SMEs)[1], [2]. The system integrates an explainable rule engine with a machine learning component to analyze key performance metrics such as task load and deadline adherence and deliver real-time Need Help alerts alongside periodic turnover risk summaries [3]. A centralized dashboard presents these insights to support consistent, evidence-based workforce management.

The existing method of employee retention management, particularly within startups, is commonly derived on the basis of manual control (such as WhatsApp messages) and the occasional reviewing. These evaluation processes are time-consuming, subjective, likely to be influenced by human factor, and mostly they are not able to detect the employee at risk, until it is too late [1]. This creates a critical operational gap, where high-stress environments lead to employee burnout and costly turnover long before managers have any objective data to act upon .

The proposed platform's AI capability is based on a logical rule engine that analyses metrics such as task load and due-date adherence. This enables it to identify employees who are at high risk of experiencing burnout, and also to identify high-performing employees who are under-appreciated, recommending them for recognition. A machine learning model (Logistic Regression) is also used to forecast the overall turnover risk of each employee. The automated functionality of this platform promotes SME efficiency and speedy risk discovery, and opens up possibilities for data-driven management in resource-constrained companies[4].

To ensure this project is grounded in real-world needs, it is being developed as a case study for a specific SME client: a startup real estate tender company with approximately 5 employees. This client's environment is defined by high-stakes, project-based work with extremely tight deadlines, making it a perfect case for analyzing burnout and

performance[2]. The system processes their task data and provides the company owner (acting as Manager/HR) with real-time alerts on a dashboard. The security of the system ensures that sensitive employee data is only visible to authorized roles.

Finally, this project would fill the gap between manual, subjective management and modern, data-driven analytics of the SMEs. This platform can overcome the human constraints by adopting an automated monitoring system, which achieves the task of decreasing turnover by means of proactive intervention. It proves that AI technologies are available to be successfully implemented to secure the financial stability as well as the most useful assets of the startup people.

1.2 Problem Statement

A key challenge in Small and Medium Enterprises (SMEs) is employee retention, especially in high-stress, project-based industries such as the real estate tender industry. Skilled employees are essential in such an environment, yet the well-being of such personnel is usually managed manually, through informal means of communication like WhatsApp, and through subjective year-end evaluations. The result of this strategy is delayed response, subjective evaluation, and unequal recognition[1], [2].

Lack of an automated predictive system is a contributing factor to high turnover, as at-risk employees are not usually identified. This subjects the business to high replacement expenses, which studies estimate to be over 33% of an employee's annual salary an expense most SMEs cannot afford[5]. Thus, the inability to proactively deal with burnout and recognition presents a direct risk to the financial prosperity and development of the SME.

This issue is magnified in startups that are dependent on decentralized, manual tools. A centralized and real-time performance and well-being monitoring system is completely lacking. Managers are not able to efficiently monitor workload balance across various projects, track performance patterns, and provide objective, evidence-based recognition.

As a result, no systematic deterrent against burnout exists, and the company culture of appreciation is compromised.

To address these weaknesses, this project proposes the AI-Driven Employee Retention Platform, a web system developed as a case study for a startup tender company. It applies a rule-based engine and a machine learning model to analyze key performance indicators, including task load and due-date adherence. The system will reduce the dependence on manual supervision, increase management objectivity, and improve employee retention by providing automatic Need Help detection, proactive recognition alerts, and a well-organized turnover risk history [4][3].

1.3 Objectives

- To analyze existing SME management systems and identify the key performance indicators (KPIs) correlated with employee Need Help and Turnover.
- To design and develop an AI-driven platform that integrates a rule-based engine for Need Help detection and recognition recommendations, and a machine learning model (Logistic Regression) for turnover-risk prediction, to proactively analyze employee performance data.
- To evaluate the predictive accuracy of the AI models and validate the platform's overall effectiveness and usability through User Acceptance Testing (UAT).

1.4 Scope

System Scope:

This project will deliver a web-based platform, the AI-Driven Employee Retention Platform, designed as a case study for a Small to Medium Enterprise (SME). The system is specifically scoped for a startup real estate tender company with approximately 5 employees and will not include complex ERP modules such as accounting, CRM, or inventory. The system will include core modules for Task Management (allowing managers to assign Task Load and Due Dates), Performance Ratings, and Leave Management. It will use a Rule-Based AI Engine to analyze real-time performance metrics to generate proactive Need Help alerts and recognition suggestions. The system will also use a Machine Learning Model to predict a periodic Turnover Risk Score for each employee. A centralized dashboard will present these alerts and reports, managed through a secure, role-based login system.

Development Scope:

The backend for the system will be developed using Node.js (with the Express framework). The frontend interface will be developed using React.js. The Rule-Based AI (for Need Help /recognition) will be developed as part of the Node.js backend. The AI prediction model (for turnover risk) will be developed using Python (scikit-learn), applying a Logistic Regression algorithm trained on a synthetic CSV dataset. The database for the system will be PostgreSQL. Visual Studio Code will be the main IDE used for all coding.

User Scope:

The system involves three user roles. The **Admin/HR** manages the system, employee accounts, and views all company-wide AI reports. The **Manager** assigns tasks, gives performance ratings, and views AI alerts for their specific team. The **Employee** views and completes their assigned tasks, submits leave requests, and views their personal dashboard.

CHAPTER 2

2.1 Introduction

This chapter analyses the literature that is associated with the AI-Driven Employee Retention Platform. It looks at the current status of SMEs in terms of performance and employee Need Help and turnover, and seeks to know what works and what does not work in the current approaches, as well as the obvious areas of gaps in the current approaches [1][2]. The review initially outlines the most important concepts in employee retention, Need Help and HR analytics and summarizes the literature that assumes a rule-based methods and machine learning models to predict attrition or identify high-risk employees [3]. It then lists some current ERP and HR systems available to SMEs, including odoo, orange HRM and ERPnext and specifically looks at the features of these systems in terms of employee-management and performance-monitoring aspects[6], [7], [8]. By comparing the strengths and weaknesses of these systems and analysing how they support (or fail to support) proactive retention, this chapter highlights the limitations of current manual and system-based practices. This review results can be the basis of the creation of a new platform that offers real-time Need Help alerts and turnover-risk predictions based on a small startup tender company.

2.2 Existing Systems

2.2.1 Odoo

Odoo is an open-source, modular ERP solution that is very popular among SMEs as a tool to streamline other business processes, human resource being one of them [6]. As part of its HR suite, Odoo has an employee records management module, staff attendance management, time-off, employee appraisals, and employee organizational structure management. It enables organizations to develop structured performance appraisal and track turnover with the help of the Employee Analysis report that shows trends in turnover and retention over the time [6]. Also, Odoo allows setting goals, 360 feedback, and custom evaluation templates, which makes them useful in structured performance management. These attributes make the employees stay as they spot their strengths, weaknesses and areas of growth.

Even though Odoo facilitates intuitive reporting, it does not have in-built AI-powered burnout or real-time predictive retention analytics. Certain predictive applications such as turnover analysis can be found on the Odoo App Store as a third party module [9]. Such add-ons provide simple predictive algorithms, including machine learning or logistic regression, to estimate the probability of employee turnover. However, such tools are not deeply embedded or explainable for SME-specific use cases. On the one hand, due to the high level of customization, Odoo can be integrated with numerous applications; on the other hand, these sophisticated HR analytics demand technical skills or external systems [6], [9]. Accordingly, Odoo offers a robust administrative/reporting base and fails to offer proactive, AI-based retention assistance.

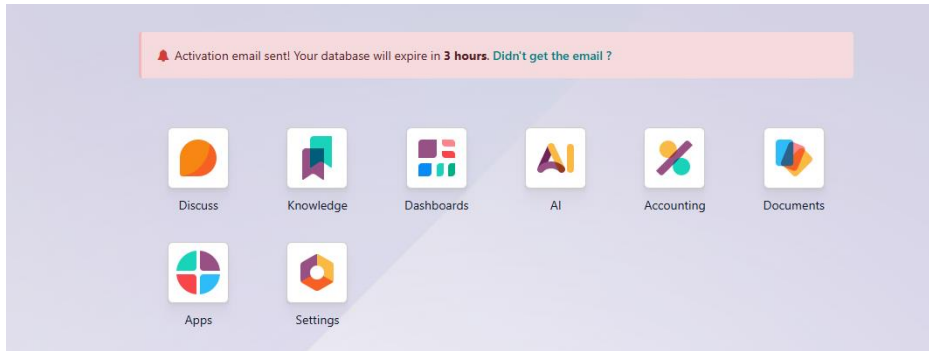


Figure 2.1 Odoo main system plugins

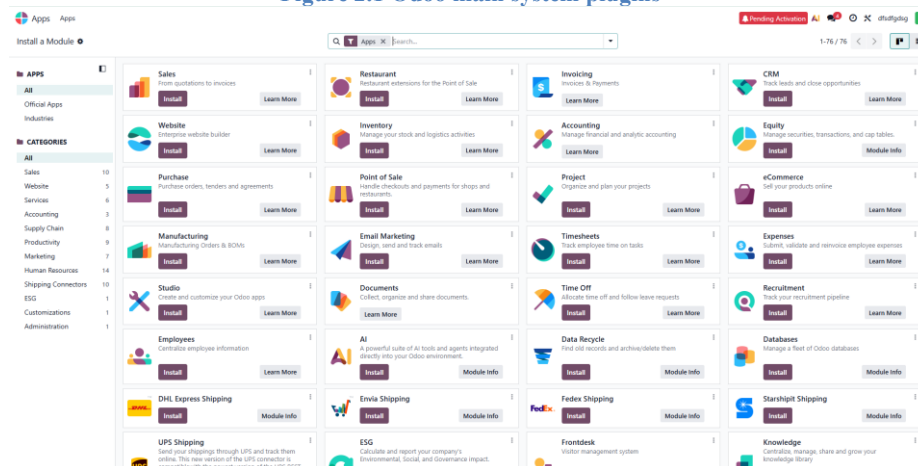


Figure 2.2 Odoo install plugins page

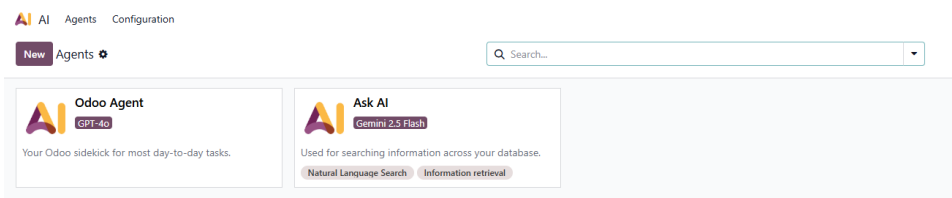


Figure 2.3 Odoo AI plugins

Figure 2.1, 2.2 and 2.3 shows interface of Odoo platform and its modular design in form of apps (plug-ins), which in this case are dashboards, documents, accounting and settings. It also demonstrates the AI section of Odoo that supports configurable AI agents that can serve the user by retrieving data in the system database and responding to the query in natural language. These AI plug-ins are an in-built assistant to enhance productivity, as they allow quicker search, summarizing, and accessing information in the Odoo environment.

2.2.2 OrangeHRM

OrangeHRM is a dedicated HR management platform known for its user-friendly interface and targeted functionality for small and medium enterprises [7]. The system includes core modules for employee information management, recruitment, leave and attendance tracking modules, performance appraisal module, and simple HR analytics module. It offers the tools to track the employee turnover, engagement, and structured appraisal, which are all necessary to understand and enhance the employee retention [7]. Evaluation forms help managers to spot the best and poor employees, which inform their decisions about the promotion and training. The workflows can also be configured and user-style access to the system provides secrecy to sensitive evaluations.

OrangeHRM has already made a step in the direction of the artificial intelligence implementation with the help of the functions like the AI-driven performance summaries and automated assessment recommendations. In its business version, it also brings predictive analytics and virtual HR assistants which enhance the efficiency of the HR workflows. However, these tools are not focused on real-time burnout detection or dynamic turnover-risk alerts. In addition, the majority of these AI capabilities are available only in paid versions, and many SMEs cannot afford it without the flexibility of the budget [7]. In essence, while OrangeHRM offers practical tools for managing HR operations and basic analytics for retention, its predictive capabilities and AI features are limited and not explicitly tailored to small, high-pressure environments.

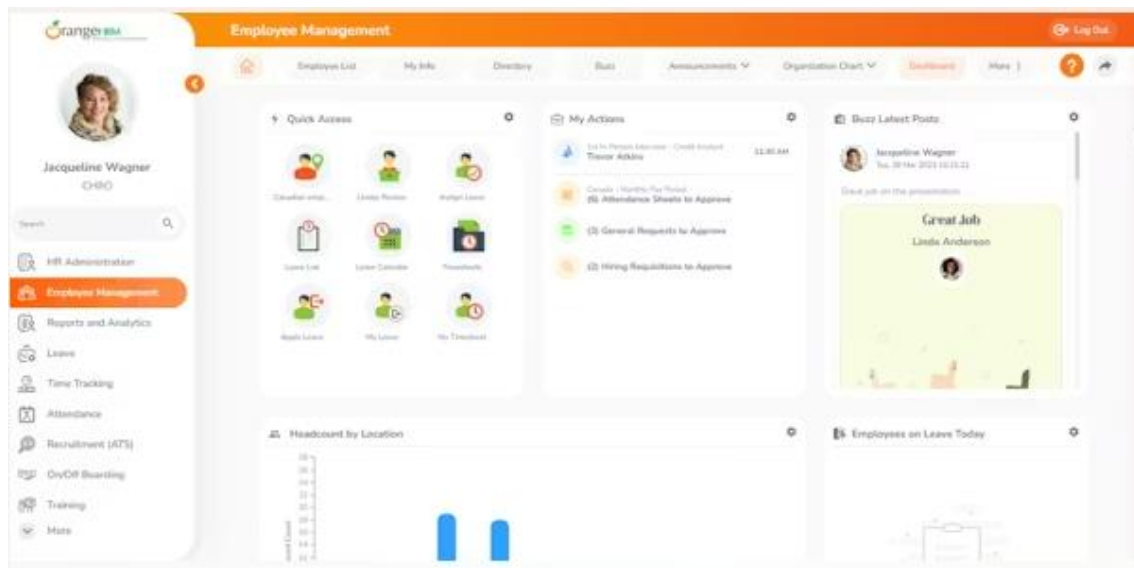


Figure 2.4 OrangeHRM

The dashboard interface of the OrangeHRM Employee Management is presented in the figure 2.4. It has a left-side navigation menu that allows entry to the key HR modules, including employee administration, time tracking, leave management, recruitment, and reports/analytics. The primary dashboard area includes the quick access shortcuts, task/action list (e.g., approvals or pending items), and summary widgets that allow the HR staff to see the workforce activity (e.g., the latest updates, employee status, and simple charts). In general, the design emphasizes the attention of OrangeHRM to centralized HR activities and a clean and role-based dashboard to manage and monitor day-to-day activities.

2.2.3 ERPNext

ERPNext is a full-fledged ERP software with a complete HR module, which is open based on open-source[8] . It handles the core processes that include recruitment process, payment process, performance appraisals, and separation of employees. The system also captures the resignation causes and exit interviews, which aid the retention analysis using the descriptive statistics. ERPNext can also be used to manage the performance because it allows the managers to allocate employees and monitor their achievement and have organized appraisals. These tools can give an insight on strengths and gaps in the workforce that can be used to develop and retain employees.

ERPNext does not have inbuilt AI or intuitive burnout detection systems, despite its flexibility and high level of integration with other modules of businesses [8]. Although retention analysis may be offered with custom machine learning models as proposed by the developers of the community user base, they are not part of the official release and would need to be developed in-house or with the help of third-party tools. However, ERPNext has an open-source nature that can be customized, but this requires technical expertise, which is not always present in SMEs. Thus, despite the fact that ERPNext can be used to facilitate well-organized HR processes and fundamental retention analysis, it is not as predictive, capable of identifying risks in real-time, or automatically recognizing employees [8].

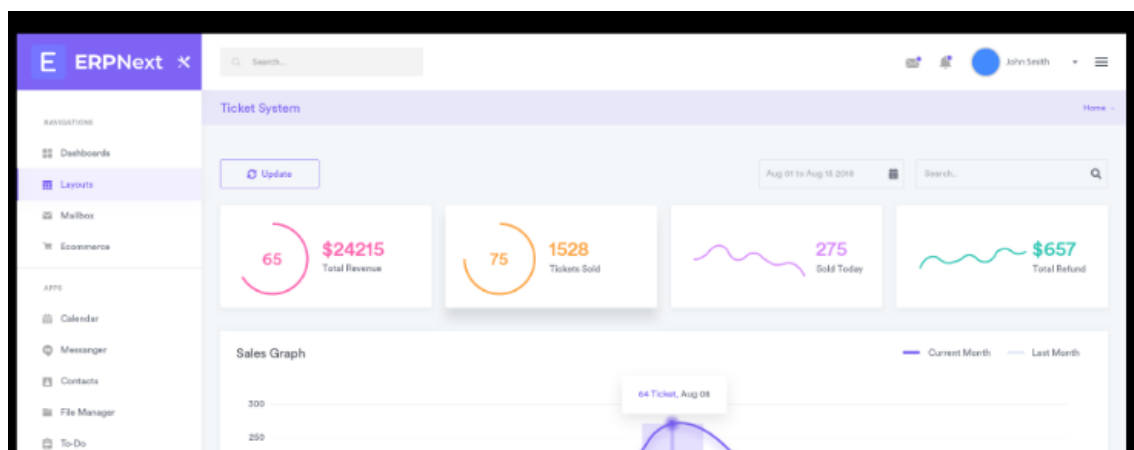


Figure 2.5 ERPNext Layout

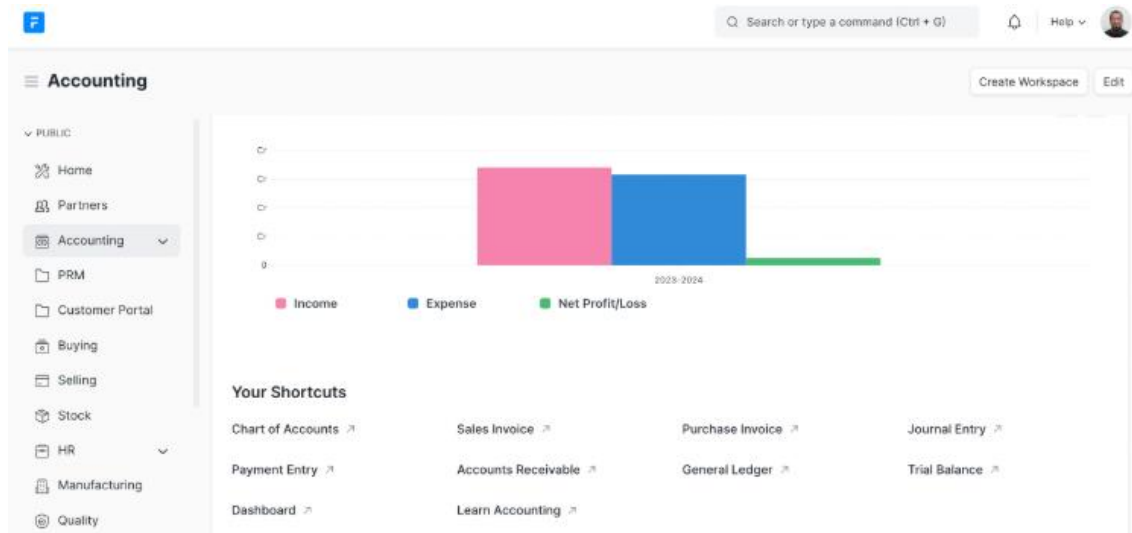


Figure 2.6 ERPNext account function

These Figure 2.5 and 2.6 demonstrate the clean ERP-style dashboard layout of ERPNext with a sidebar menu of modules and a central workspace that displays charts, shortcuts and summary cards. The interface also focuses on fast navigation, real time reporting graphics, and role-based access to various business operations, which is common with ERP systems that are meant to be used in well-organized management and monitoring.

2.3 Comparison of Existing System

Table 2.1 Comparison of Existing System

Comparison Attributes	Odoo	OrangeHRM	ERPNext	Proposed System
Description	A modular open-source ERP system with HR tools for SMEs.	An HR-specific platform with SME-focused modules and user-friendly UI.	A full open-source ERP with HR capabilities for SMEs.	A platform focused on employee retention using real-time Need Help alerts and predictive turnover scoring.
Features	Employee records, appraisals, goal tracking, 360 feedback, reporting.	Leave management, attendance, appraisals, reporting, some AI tools in paid versions.	Recruitment, payroll, appraisals, exit interviews, custom surveys.	Task tracking, rule-based burnout detection, logistic regression for turnover prediction, recognition alerts.
Strengths	Highly customizable, integrates with business functions, supports structured appraisals.	Good usability, modular design, supports structured reviews and engagement tracking.	Flexible and low-cost, integrates well with ERP modules, open-source customization.	Built-in AI and rule engine, real-time alerts, SME-targeted dashboard, proactive retention focus.
Weaknesses	No built-in AI for burnout/retention prediction, needs external modules.	AI tools limited to commercial edition, no real-time	Lacks native AI/burnout detection, requires in-house	Still under development; limited scope

		time burnout alerts.	development for analytics.	beyond HR analytics.
Advantages	Streamlines HR workflows and performance tracking effectively.	Easy to use, structured HR features, promotes basic analytics.	Supports standard HR operations and retention tracking via exit data.	Tailored to small teams, improves decision-making using explainable AI.
Disadvantages	Requires technical skills for advanced features; AI support depends on 3rd-party apps.	Limited predictive tools, not fully AI-integrated for SME use cases.	No AI or automated alerts by default, limited analysis tools.	Not a full ERP, requires proper data collection to function effectively.

Table 2.1 is a comparison of three systems that are currently implemented: Odoo, OrangeHRM, and ERPNext, with the proposed AI-Driven Employee Retention Platform. Odoo is a generalizable ERP platform that has organized HR features like appraisals and reporting, but does not have inbuilt AI to detect burnout proactively or predictive turnover. Although it also accommodates the third-party predictive modules, they also demand technical knowledge and are not smoothly incorporated in the circumstances of SMEs. OrangeHRM is easy to use and has a convenient HR interface, especially designed to meet the needs of SMEs and incorporates performance assessment and some AI functionalities such as summary of appraisals. Nevertheless, its more sophisticated functions are confined to the commercial edition and real time risk detection functionalities are not in-built. ERPNext is a general, open-source ERP platform, with well-integrated HR modules, such as appraisals and exit interview tracking, but does not provide native AI or employee risk analytics automation. In comparison, the suggested system directly fills these gaps by integrating rule-based engine and machine learning algorithms in the specific context of SMEs, providing real-time burnout notifications, turnover forecasts, and recognition recommendations via a specialized, explainable, and lightweight decision-support dashboard.

2.3.1 Relevance of Comparison with Project Title

The applicability of this comparison can be attributed to the fact that it has identified constraints in the existing HR and ERP systems that SMEs use to manage and retain employees [1], [2]. Although Odoo has a high-quality HR administration system and organised appraisals, it does not have native AI-based features to identify burnout or risk of turnover in real-time. This restricts its capability to facilitate proactive retention measures within the high-paced SME context [6], [9]. OrangeHRM offers a dedicated HR solution, which has certain features that are enhanced with AI, like automated feedback summaries, but the majority of advanced analytics are also available only in a paid version, which is not accessible to startups with limited budgets [7]. ERPNext is highly versatile, yet affordable, but does not have built-in AI capabilities and real-time analytics, so it would need technical development to provide predictive retention features [8].

The AI-Powered Employee Retention Platform offered in the proposal directly fills these gaps, integrating the burnout detection rules with machine learning turnover prediction designed with SMEs in mind [3]. In contrast to general-purpose ERPs, this one is specifically designed to track the amount of tasks assigned to employees and their deadlines met, send real-time burnout warnings, as well as make recognition recommendations to high-performing but underappreciated employees. These are essential in small teams where individual performance of the employees has a great influence on the whole operations. In addition, the dashboard of the proposed platform is simplified and can be used by non-technical HR-related individuals or startup founders, so it is incredibly approachable and easy to take action. This will not only ensure that decisions about retention are based on data rather than intuition but also enable SMEs to act before the problem of burnout or a turnover is too late. The platform extends beyond what the current systems can provide by concentrating on the unique retention issues of small, high-stress workplaces, and provides a smart, niche solution to a long-standing issue [4].

2.4 Summary

As the comparison in Table 2.1 demonstrates, the existing HR and ERP systems, Odoo, OrangeHRM and ERPNext, have their advantages and disadvantages when addressing the issue of employee retention in SMEs[1]. Although both systems facilitate key HR processes like performance reviews and simple analytics, they do not provide AI-driven insights, live burnout notifications, and predictive turnover forecasts [3]. The absence of integrated and explainable tools that are small, high-pressure team specific means there is an evident gap in their capability to actively address staff health.

In summary, the suggested AI-Driven Employee Retention Platform will help to overcome these limitations by integrating both rule-based burnout detection with the machine learning-based turnover prediction specifically designed to meet SME-specific requirements [4]. Its simplified, easy to use dashboard allows its managers to make prompt and informed decisions, which will enhance employee recognition, decreasing the turnover rate, and increasing overall stability of the workforce. This solution will enable startups to leave behind manual oversight and reactive human resources instead of providing a contemporary, data-driven method to the employee retention issue.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter represents the general methodology of the AI-Driven Employee Retention Platform development which includes the combination of both the project-based and research-based approaches. A functional web system will need to be developed in the project, whereas the research aspect will be aimed at determining the presence of KPIs, designing Need Help-detection rules, and developing a machine learning model for turnover prediction [3].

In order to sustain these changing requirements, the Agile Software Development Life Cycle (Agile SDLC) was implemented [10], [11]. Its iterative format enables features of the system to be constantly improved, along with user interface design and AI elements, depending on test and stakeholder feedback. Agile is particularly appropriate in the field of AI-related projects where models and rules need to be checked and modified on a regular basis.

This chapter explains the approach that was utilized in the course of the project, such as the project management framework, system and user requirements, proposed system design, data handling approach, AI model development, and testing strategies used to validate the performance and functionality of the platform [4].

3.2 Project Management Framework / Methodology

3.2.1 Agile Software Development Life Cycle (Agile SDLC).

The AI-Driven Employee Retention Platform is developed according to Agile SDLC, a loose software project management methodology that is actively utilized in the creation of modern software systems that need constant improvement [10]. Agile is appropriate in this project since the platform combines various dynamic elements like rule-based Need Help detection engine, logistic regression prediction model, and interactive dashboard, which demand the process of retesting and correcting.

Agile focuses on short developments (sprints), regular feedback and incremental delivery. These features make sure that the features of the system and the artificial intelligence can be enhanced in the future according to the actual testing outcomes, the feedback given by the user during the SME case study, and the verification of the performance indicators (KPIs).

Agile method also facilitates quick prototyping of modules like task management, performance rating, turnover prediction and visualization of alerts as well as assuring that every cycle results in a working and testable component of the system [11].



Figure 3.1 Agile Software Development Life Cycle

Agile SDLC Phases and Activities

1. Requirements Phase

The phase establishes the basis of the AI-Driven Employee Retention Platform by formulating the problem, requirements, and clear priorities on which the development should focus. The SME case study (manager and staff) was used to gather requirements in order to identify the main challenges faced by the company, including Need Help, imbalanced workload, and employee recognition. According to these requirements, the primary modules were validated (Task Management, Ratings, Leave, AI Alerts, and Dashboard) and the development tools were identified (Node.js, React.js, PostgreSQL, and Python). The most important KPIs were also determined to drive the AI components in the identification of Need Help and prediction of turnover. Lastly, a first sprint backlog and schedule were made to plan the work and facilitate iterative delivery [10].

2. Design Phase

This step is aimed at establishing the general design of the AI-Driven Employee Retention Platform, as well as the interaction between each element. The system architecture **Figure 3.2** will be such that the frontend web dashboard will be linked to the Node.js/Express backend API, which will be linked to the PSQL database to store users, projects, tasks, alerts, and leave requests. Moreover, the backend is connected to the rule-based engine (to receive Need Help and recognition alerts), Python logistic regression service (to predict the turnover risk). In this stage, the preparation of key diagrams, including context diagram, use case diagram, and ERD are carried out and the database schema is defined to support all the necessary modules. This makes the platform design understandable, scalable and development and integration ready [11].

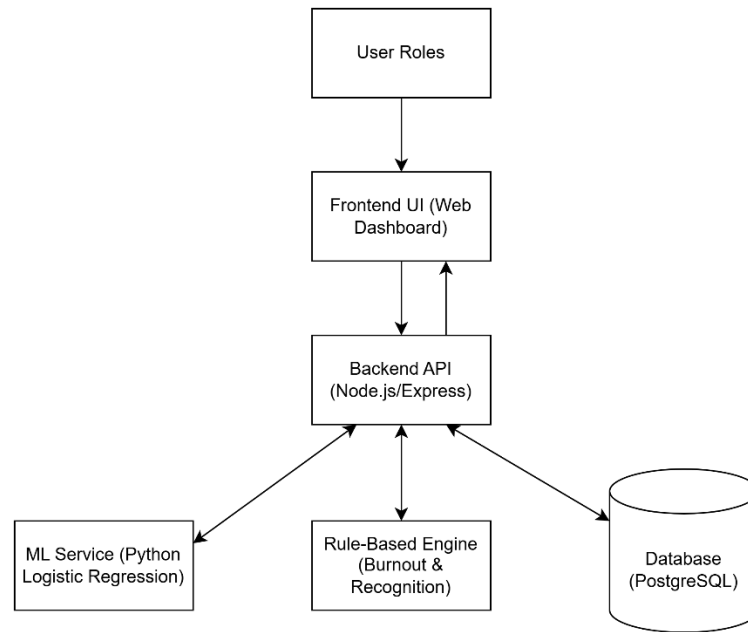


Figure 3.2 Design Structure

2.1 AI Algorithms and Measurement Approach

The subsection explains the AI methods applied to the AI-Driven Employee Retention Platform and how each alert/prediction is created. The platform is based on two methods: (1) an explainable rule-based engine to Need Help and recognition alerts, and (2) a machine learning model (Logistic Regression) to predict turnover-risk. The AI outputs are all based on KPIs gathered within the system modules (projects, tasks, ratings, and leave requests).

2.1.1 Rule-Based Engine (Need Help and Recognition)

Purpose: Issue clear, real-time notifications, which can be comprehended and taken action by managers and Admin/HR.

Mechanism: The engine constantly computes the employee KPIs within a chosen time frame (e.g., last few weeks). It then uses IF-THEN rules to issue alerts. The alerts have a clear reason that indicates the KPI conditions that triggered the alert.

A) Need Help Alerts

When the KPIs show workload pressure and deadline difficulty, Need Help alerts are generated.

Key KPIs used:

- Counting overdue work (repeated overdue work)
- Pattern of workload (frequent Heavy workload)
- Pattern of task completion (late completion pattern / recurring delays)
- Frequency of leave requests (frequency of leave requests as a stress indicator)

Output: Need Help alarm (reason and level of severity).

B) Recognition Alerts

When KPIs indicate high performance, recognition alerts are created.

Key KPIs used:

- Task ratings obtained (high average rating / repeated strong ratings)
- Punctuality (meeting deadlines on a regular basis)
- Accomplished work (consistent productivity)

Output: Recognition recommendation alert (with reason).

2.1.2 Machine Learning Model (Turnover Prediction — Logistic Regression)

Purpose: Use KPI patterns to estimate employee turnover risk and provide an early warning to Admin/HR.

Model: Logistic Regression, which gives a probability score ranging between 0 and 1.

Input KPIs:

- Overdue tasks count
- Light/Medium/Heavy distribution of workload.
- Pattern of task completion (late trend vs on-time trend)
- Ratings obtained on tasks (low ratings trend / falling average)
- Frequency/count of leave requests.
- Completed tasks count

How prediction works:

1. The system sums up the KPI values of each employee into a summary (feature set).
2. The result of the Logistic Regression is a turnover probability $P(\text{turnover})$.
3. Decision rule: when $P(\text{turnover})$ exceeds 0.70, the system sends a Turnover Risk Alert.

3. Development Phase

During this stage, the AI-Driven Employee Retention Platform is created in a series of Agile sprints, with each sprint bringing a workable module or enhancement that can be tested shortly. The development work involves the deployment of the backend APIs to facilitate the work and the performance ratings and the KPI data processing and the construction of the rule-based Need Help and recognition engine depending on the set KPI thresholds. Parallel to this, the Logistic Regression turnover prediction model is trained on the ready data, and it is exported to fit into the system. Frontend applications like the task management application, the rating application and the dashboard are built and linked to the backend APIs to maintain a complete functionality. Lastly, authentication and role-based access control are provided to protect the platform and make sure that every user role can access only the features and data that they are allowed to use [10].

4. Testing Phase

This step is used to make sure that every feature of the system is functioning properly and that it matches the expectations of the user and in Agile it is done on a continuous basis throughout each sprint in order to keep the quality. Functional and unit testing of all modules will be done to ensure that all important features like managing the tasks, ratings, and leave requests are functioning as intended. The rule-based Need Help and recognition notifications are debugged against sample KPI scenarios to ensure that the logic generates realistic results. The machine learning turnover prediction model is also tested by verifying the prediction performance and accuracy of the model using the evaluation results. Besides this, User Acceptance Testing (UAT) is done among the SME users to ensure the usability and feasibility in the actual case study environment. All problems that are identified are corrected and enhancements made and then the next sprint is proceeded to [12].

5. Deployment Phase

During this stage, the AI-Driven Employee Retention Platform is implemented into a live or virtual environment to allow users to access and test it. As Agile allows continuous improvement, it can be deployed more than once as new features are added or refined between sprints. The database and the backend services are migrated to a cloud hosting environment, and the necessary deployment settings including environment setup, hosting setup and security settings are implemented. Upon deployment, internal SME testing of the system is done to ensure that the major functions are functioning properly under real usage conditions. Lastly, the platform behaviour and performance is checked end to end using real users to verify that it is stable, properly data handling and dashboard updates [10].

6. Review Phase

At the end of each sprint, a review is carried out to assess the completed work and collect feedback for continuous improvement. The sprint output is evaluated with the project supervisor and the SME manager to confirm that the delivered features meet the intended requirements. Usability aspects such as system navigation, dashboard clarity, and alert accuracy are also checked to ensure the platform remains practical for real use. Based on the feedback, KPI thresholds may be adjusted and AI-related behaviour refined to improve the reliability of insights. Finally, the priorities are updated, and new improvements are added to the backlog to guide the next sprint [11].

3.3 Project Requirement

3.3.1 Functional Requirements

ID	Requirement	Primary User
F1	allow users to log in and access features based on their role (Admin/HR, Manager, Employee).	All
F2	allow Admin/HR to create new users by entering name, email, role, and password.	Admin/HR
F3	allow Admin/HR to edit and delete users and set users as active/inactive if needed.	Admin/HR
F4	allow Manager to create new projects and assign a project name.	Manager
F5	allow Manager to toggle project status (Active/Done) to control visibility for employees.	Manager
F6	allow Manager to create and assign tasks under a selected project (title, description, due date, workload, assigned employee).	Manager
F7	show Manager a Team Tasks list by project with filtering (e.g., by employee, task status, workload).	Manager
F8	allow Employee to view assigned projects and open a project to see their assigned tasks.	Employee
F9	allow Employee to view task details (full description, due date, workload, status).	Employee
F10	allow Employee to update task status (To Do / In Progress / Done).	Employee

F11	allow Manager to rate completed tasks (1–5) and add comments.	Manager
F12	allow Employee to view ratings and comments received on completed tasks.	Employee
F13	allow Employee to submit leave requests (dates, type, reason).	Employee
F14	allow Admin/HR to view all leave requests and approve/reject them.	Admin/HR
F15	allow Manager to view team leave requests (view-only).	Manager
F16	display alerts and AI insights on dashboards: Admin/HR sees all alert types (Need Help, recognition, turnover).	Admin/HR
F17	display Manager dashboard alerts for Need Help and recognition only (no turnover alerts).	Manager
F18	allow Admin/HR to filter alerts by type, employee, status, and date/time.	Admin/HR
F19	allow Manager to filter team data (tasks/ratings) by employee and status.	Manager

3.3.2 Non-Functional Requirements

ID	Requirement	Primary User
NF1	enforce role-based access control so each role only sees permitted pages/features.	Security
NF2	provide a clear, professional ERP-style UI (dashboard layout, sidebar navigation, readable tables, clean typography).	Usability

NF3	be responsive and usable on common desktop browsers (Chrome/Edge) with acceptable display on tablets.	Compatibility
NF4	ensure common actions (open dashboard, list tasks/alerts) load smoothly without noticeable delay under small-SME usage.	Performance
NF5	keep data consistent (no missing links between project–task–user; no broken pages when lists are empty).	Reliability
NF6	present alerts in an understandable way (type + reason + timestamp) to support explainability.	Interpretability
NF7	maintain simple maintainability so KPI rules/thresholds and alert templates can be updated later.	Maintainability

3.3.3 Constraints and Limitations

ID	Constraint / Limitation	Primary User
C1	the study is based on one SME case study (≈5 employees).	limited generalisation
C2	real HR data is sensitive , so turnover model training may rely on synthetic/limited data .	affects model realism
C3	AI outputs are represented as sample alerts in the current prototype stage (not fully automated live predictions).	prototype scope
C4	system scope focuses on projects, tasks, ratings, leave, and alerts only (not payroll, recruitment, full HR suite).	scope boundary
C5	predictions/alerts are decision support , not a guarantee of actual turnover or Need Help.	ethical/accuracy boundary

3.4 Propose Design

3.4.1 Context Diagram

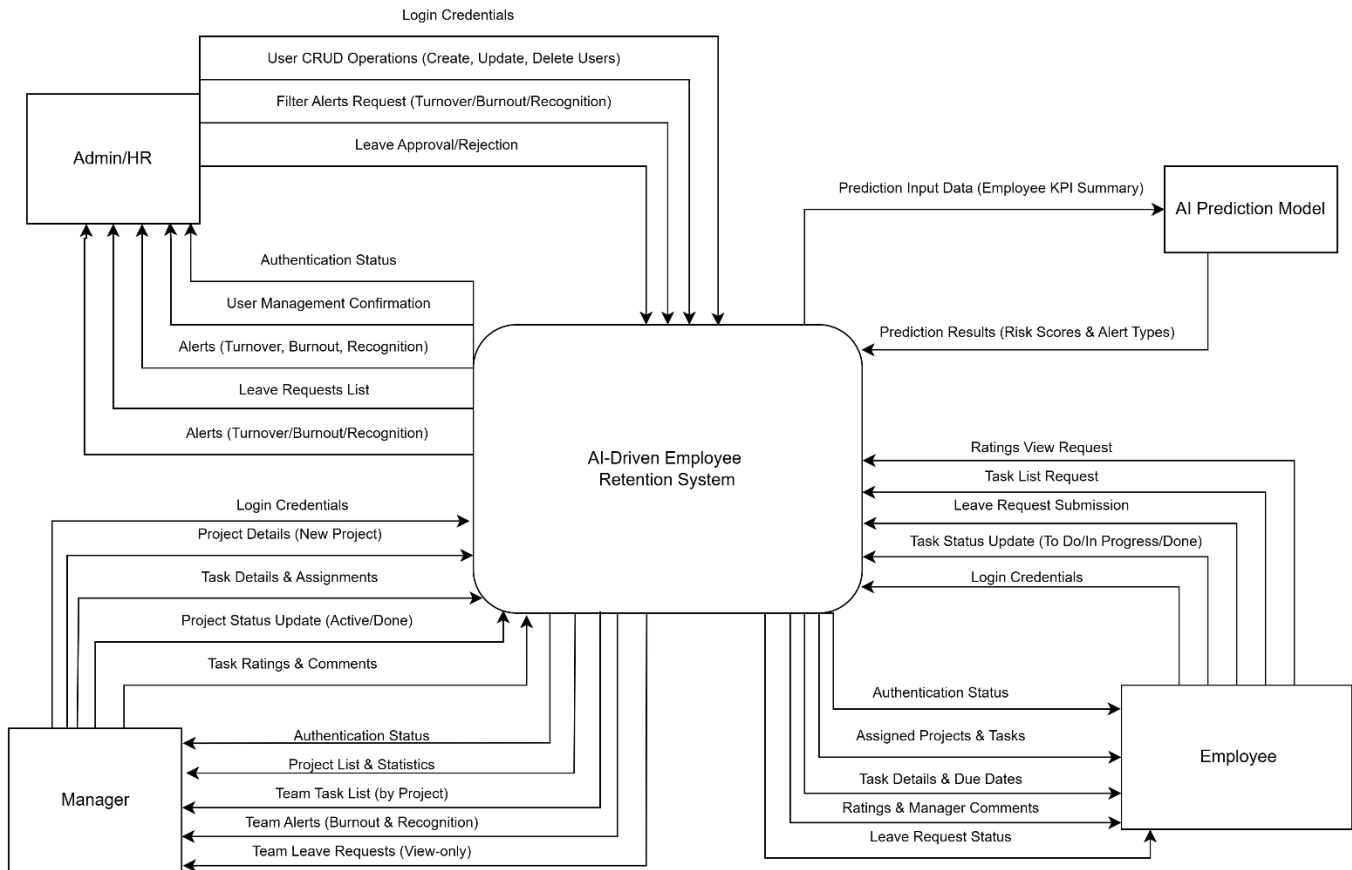


Figure 3.3 Context Diagram

Figure 3.3 shows the context diagram for the **AI-Driven Employee Retention Platform**. The diagram presents the high-level interaction between the system and its external users, which include **Admin/HR**, **Manager**, and **Employee**, as well as the external **AI Prediction Model**. Admin/HR logs into the system to perform user management (create, update, delete users), approve or reject leave requests, and monitor all alert types (turnover, Need Help, and recognition) with filtering options. The Manager logs in to create projects, assign tasks to employees, update project status (active/done), and rate completed tasks, while also viewing team-related alerts (Need Help and recognition only) and team leave requests for awareness. Employees interact with the platform by logging

in, viewing assigned projects and tasks, updating task progress status (to do/in progress/done), submitting leave requests, and viewing ratings and comments provided by the manager. In the background, the system sends summarized employee KPI data (such as workload levels, overdue tasks, task completion patterns, ratings, and leave counts) to the AI Prediction Model and receives prediction results in return, which are then presented to users as actionable alerts and insights through the dashboard.

3.4.2 Use Case Diagram

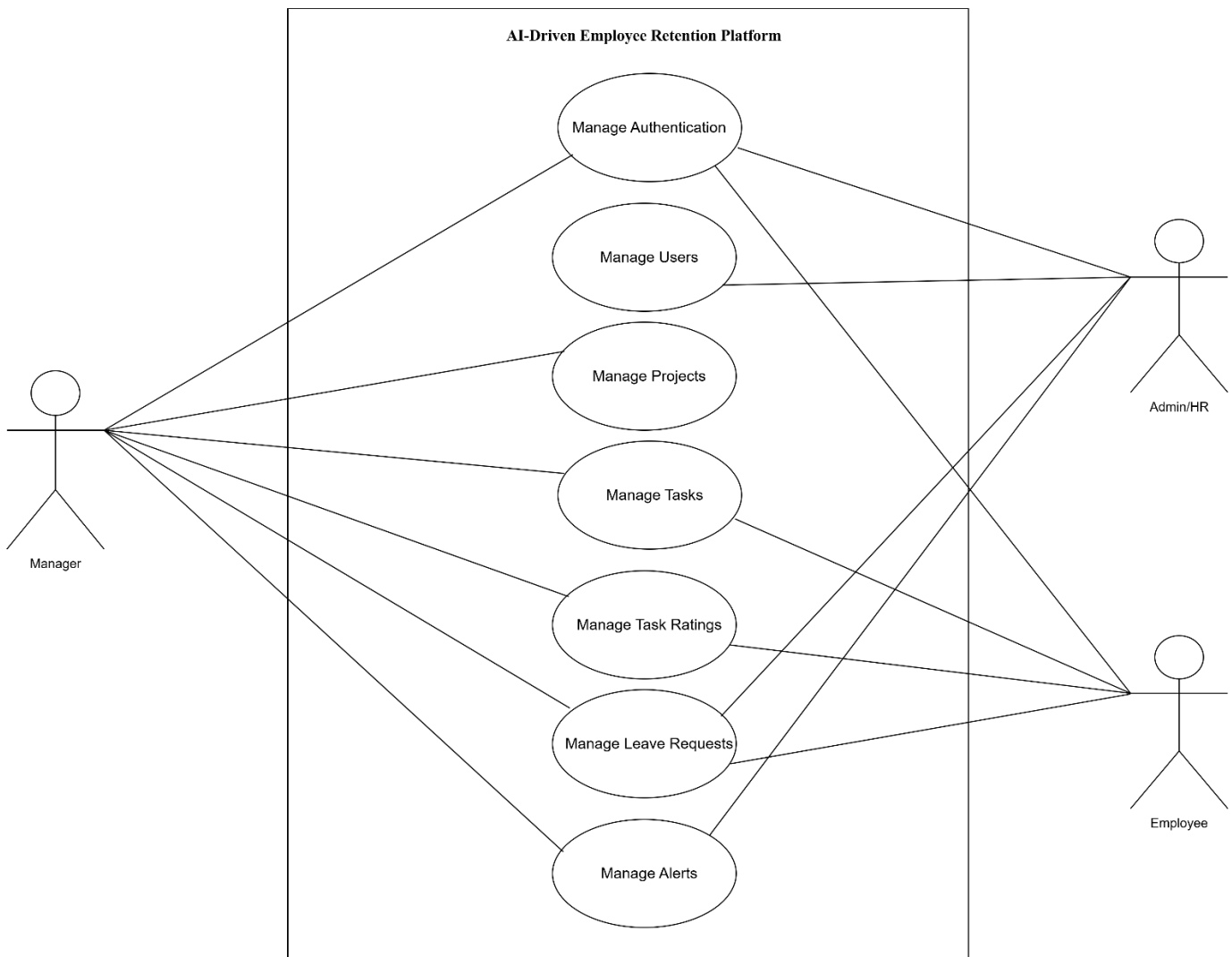


Figure 3.4 Use Case Diagram

The use case diagram illustrates the main functions of the **AI-Driven Employee Retention Platform** and how each user role interacts with them. **Admin/HR** accesses the system to manage user accounts, handle leave requests, and monitor all alerts. The **Manager** manages projects and tasks, rates completed tasks, views team leave requests, and monitors relevant alerts. The **Employee** interacts with the system mainly to manage assigned tasks, submit leave requests, and view task ratings. All roles require authentication to access the platform features.

3.4.3 Use Case Description

Table 3.1 Manage Authentication

Field	Description
Use Case ID	UC01
Use Case Name	Manage Authentication
Actor	Admin/HR, Manager, Employee
Pre-condition	User account exists, is active, and user has valid email and password.
Basic Flow	1. User opens the login page. 2. User enters email and password. 3. System verifies the credentials. 4. System identifies the user role (Admin/HR or Manager or Employee). System redirects the user to the correct dashboard based on role.
Post-conditions	User is logged in and can access features allowed for their role.
Rules	Access is role-based: • Admin/HR can access user management, all alerts, and leave approvals. • Manager can access projects, tasks, task ratings, team alerts (Need Help & recognition), and team leave view-only. • Employee can access assigned tasks, task status updates, ratings view, and leave request submission.
Constraints	Login fails if credentials are incorrect or the account is inactive.

Table 3.2 Manage Users

Field	Description
Use Case ID	UC02
Use Case Name	Manage Users
Actor	Admin/HR
Pre-condition	Admin/HR is logged in and has permission to manage users.
Basic Flow	1. Admin/HR opens the User Management page. 2. System displays the list of users (name, email, role, status). 3. Admin/HR selects an action: Add / Edit / Delete / Activate-Deactivate. 4. Admin/HR enters or updates user details (name, email, role, password) and confirms. 5. System validates the input (e.g., unique email). 6. System saves the changes and shows a confirmation message.
Post-conditions	User record is created/updated/deleted, and the updated user list is available in the system.
Rules	• Only Admin/HR can manage users. • Email must be unique.
Constraints	Cannot create a user without required fields (name, email, role, password).

Table 3.3 Manage Projects

Field	Description
Use Case ID	UC03
Use Case Name	Manage Projects
Actor	Manager
Pre-condition	Manager is logged in and has permission to manage projects.
Basic Flow	1. Manager opens the Projects page. 2. System displays the project list (project name, status, basic statistics). 3. Manager selects Create Project. 4. Manager enters the project name and confirms. 5. System validates the input and saves the new project as Active. 6. Manager can change a project status by selecting Toggle Status (Active ↔ Done). 7. System updates the status and refreshes the project list.
Post-conditions	A project is created or updated (status changed), and the updated project list is available for task assignment and employee viewing.
Rules	• Only Manager can create and update project status. • Projects have two statuses only: Active and Done. • Only Active projects should be visible/usable for employees (based on your requirement).
Constraints	Project must have a valid name. System should prevent creating projects with missing data.

Table 3.4 Manage Tasks

Field	Description
Use Case ID	UC04
Use Case Name	Manage Tasks
Actor	Manager, Employee
Pre-condition	User is logged in. At least one project exists.
Basic Flow	- Manager: 1. Manager opens Tasks and selects a project (Active/Done). 2. System shows the Team Tasks table with filters (employee/status). 3. Manager clicks Create New Task, fills the form, and submits. 4. Manager enters the project name and confirms. 5. System saves the task, shows success, and returns to Tasks. - Employee: 1. Employee opens My Tasks and selects a project. 2. System shows the employee's assigned tasks for that project. 3. Employee opens Task Details (View More), updates status, and submits. 4. System saves the update and refreshes the task list.
Post-conditions	Tasks are created/updated, task status changes are stored, and task lists reflect the latest updates for both Manager and Employee views.
Rules	• Only Manager can create tasks and assign them to employees. • Employee can only update status for tasks assigned to them. • task statuses: todo, in_progress, done, overdue (overdue can be auto-calculated if due date passed). • Workload levels: Light, Medium, Heavy.
Constraints	-

Table 3.5 Manage Task Ratings

Field	Description
Use Case ID	UC05
Use Case Name	Manage Task Ratings
Actor	Manager, Employee
Pre-condition	User is logged in. Task exists under a project. Task status is Done (completed).
Basic Flow	- Manager: 1. Manager opens Task Ratings (or Ratings section under a selected project). 2. System displays a list/table of completed tasks that are not rated yet (with filters if available). 3. Manager selects a completed task. 4. Manager gives a rating (1–5) and writes an optional comment, then submits. 5. System saves the rating and shows confirmation. - Employee: 1. Employee opens My Ratings (or Ratings view). System shows ratings and comments given by the manager for the employee's tasks.
Post-conditions	The task rating and comment are stored and visible to the employee.
Rules	• Only the Manager can create/update ratings. • Employee can view only. • Ratings apply only to completed tasks.
Constraints	Rating cannot be submitted if the task is not marked Done.

Table 3.6 Manage Leave Requests

Field	Description
Use Case ID	UC06
Use Case Name	Manage Leave Requests
Actor	Manager, Employee, Admin/HR
Pre-condition	User is logged in. Employee account is active.
Basic Flow	- Employee: 1. Employee opens Leave Request page. 2. Employee enters leave details (start date, end date, type, reason) and submits. 3. System saves the request with status Pending and shows confirmation. - Admin/HR: 1. Admin/HR opens Leave Requests list. 2. System shows all requests with status and details. 3. Admin/HR selects a request and chooses Approve or Reject. 4. System updates the request status and shows confirmation. - Manager: 1. Manager opens Team Leave Requests. 2. System shows team leave requests for viewing only.
Alternative Flow	-
Post-conditions	Leave request is stored and its status is updated (Pending/Approved/Rejected). Employee can view the request status.
Rules	• Employee can submit leave requests. • Admin/HR can approve/reject. • Manager is view-only.
Constraints	-

Table 3.7 Manage Alerts

Field	Description
Use Case ID	UC07
Use Case Name	Manage Alerts
Actor	Manager, Admin/HR
Pre-condition	User is logged in. Alerts exist in the system (Need Help, recognition, turnover).
Basic Flow	- Admin/HR: - Admin/HR opens the Alerts page. - System displays all alert types (Need Help, Recognition, Turnover) with details (employee, reason, date/time). - Admin/HR applies filters (by type, employee, status, date) to monitor alerts. - Manager: - Manager opens the Alerts page. - System displays Need Help and Recognition alerts only for the manager's team. - Manager applies filters (by type, employee, date) to review alerts.
Alternative Flow	-
Post-conditions	Alerts are viewed and filtered based on the user's role; no alert status is updated.
Rules	- Admin/HR can view all alert types. - Manager can view Need Help & Recognition only (no Turnover). - Alerts are view-only (no marking/acknowledging in the system).
Constraints	-

3.5 Data Design

3.5.1 ERD

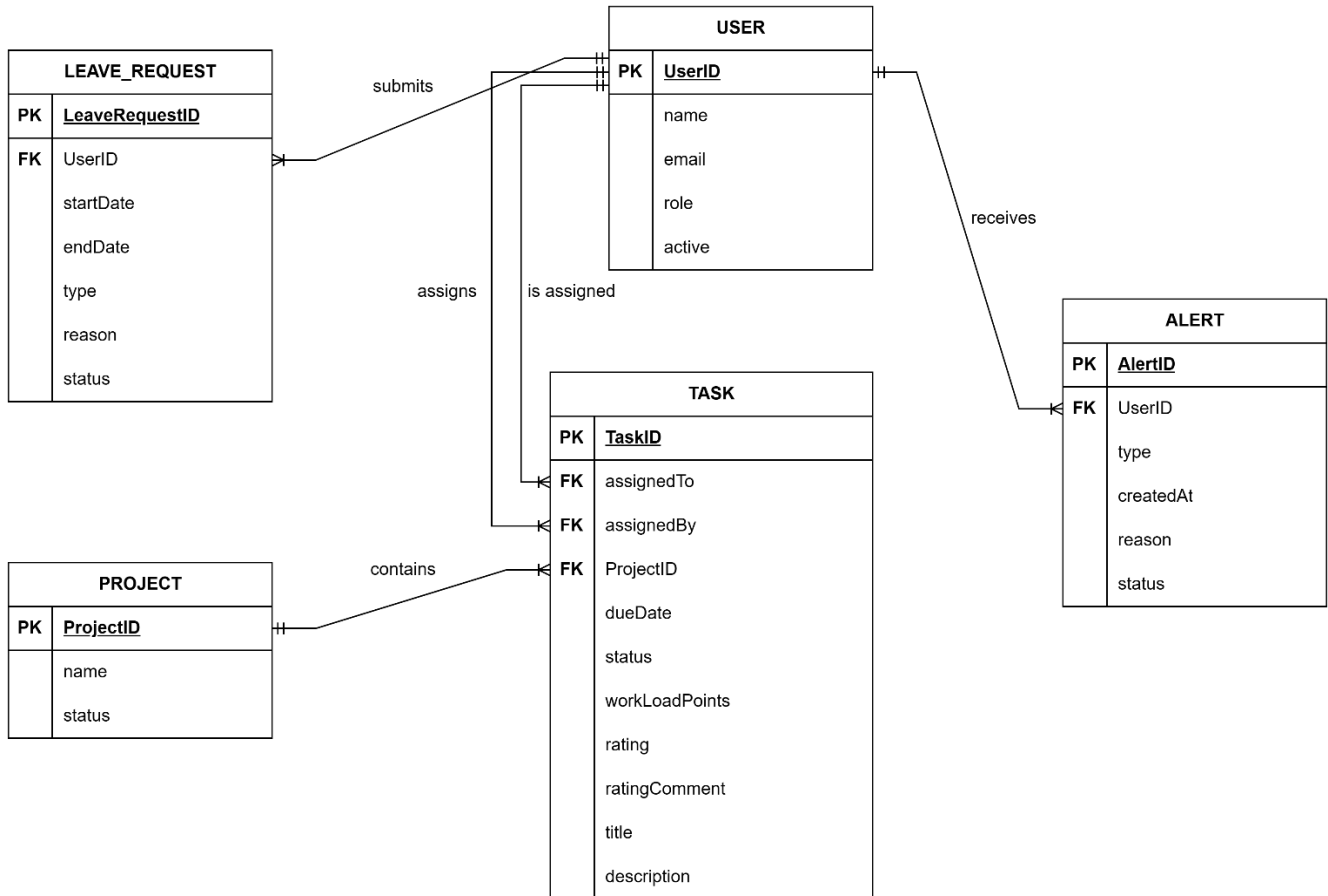


Figure 3.5 ERD

Figure 3.5 shows the Entity–Relationship Diagram (ERD) for the AI-Driven Employee Retention Platform. The USER table stores all system accounts and is related to multiple records in other tables. A PROJECT contains many TASKS, where each task is assigned by a user (manager) and assigned to a user (employee). Employees can submit multiple LEAVE_REQUESTS, and users can receive multiple ALERTS generated by the system. Overall, the ERD defines how users, projects, tasks, leave requests, and alerts are linked to support task management, monitoring, and employee retention features.

3.5.2 Database Dictionary

- USER TABLE

Table 3.8 USER TABLE

Field Name	Data Type	Key	Constraints/Values	Description
UserID	INT	PK	Auto increment	Unique identifier for each user
Name	VARCHAR			Full name of he user
email	VARCHAR		UNIQUE	Full name of the user
role	ENUM		ADMIN_HR, MANAGER, EMPLOYEE	Defines system access level
active	BOOLEAN		TURE/FALSE	Determines if the account is active

- PROJECT TABLE

Table 3.9 PROJECT TABLE

Field Name	Data Type	Key	Constraints/Values	Description
ProjectID	INT	PK	Auto increment	Unique identifier for each project
projectName	VARCHAR			Project name/titile
status	ENUM		Active, done	Project visibility/availability

- TASK TABLE

Table 3.10 TASK TABLE

Field Name	Data Type	Key	Constraints/Values	Description
TaskID	INT	PK	Auto increment	Unique identifier for each task
assignedTo	INT	FK	USER(UserID)	Employee assigned to complete the task
AssignedBy	INT	FK	USER(UserID)	Manager who created/assigned the task
ProjectID	INT	FK	PROJECT(ProjectID)	Project the task belongs to
title	VARCHAR			Task title
description	TEXT			Task details/notes

dueDate	DATE			Deadline for completing the task
status	ENUM		In_progress, done, overdue	Current task status
workLoadPoints	ENUM		Light, Medium, Heavy	Workload level set by manager
rating	INT		1-5	Rating given by manager after completion
ratingComment	TEXT			Manager feedback/comment

- ALERT TABLE

Table 3.11 ALERT TABLE

Field Name	Data Type	Key	Constraints/Values	Description
AlertID	INT	PK	Auto increment	Unique identifier for each alert
UserID	INT	FK	USER(UserID)	Employee related to the alert
type	ENUM		Turnover, Need Help, recognition	Alert category
createdAt	DATETIME			Date and time alert was created
reason	TEXT			Explanation of why the alert was generated
status	ENUM		Open, acknowledged, resolved	Alert handling status

- LEAVE_REQUEST TABLE

Table 3.12 LEAVE_REQUEST TABLE

Field Name	Data Type	Key	Constraints/Values	Description
LeaveRequestID	INT	PK	Auto Increment	Unique identifier for each leave request
UserID	INT	FK	USER(UserID)	Employee who submitted the leave request
startDate	DATE			Leave start date
endDate	DATE			Leave end date
type	ENUM		Annual, Sick, Emergency	Leave category
reason	TEXT			Reason provided by employee
status	ENUM		Pending,approved, rejected	Approval status of the request

3.6 PROOF OF INITIAL WORK

- Login page for all users

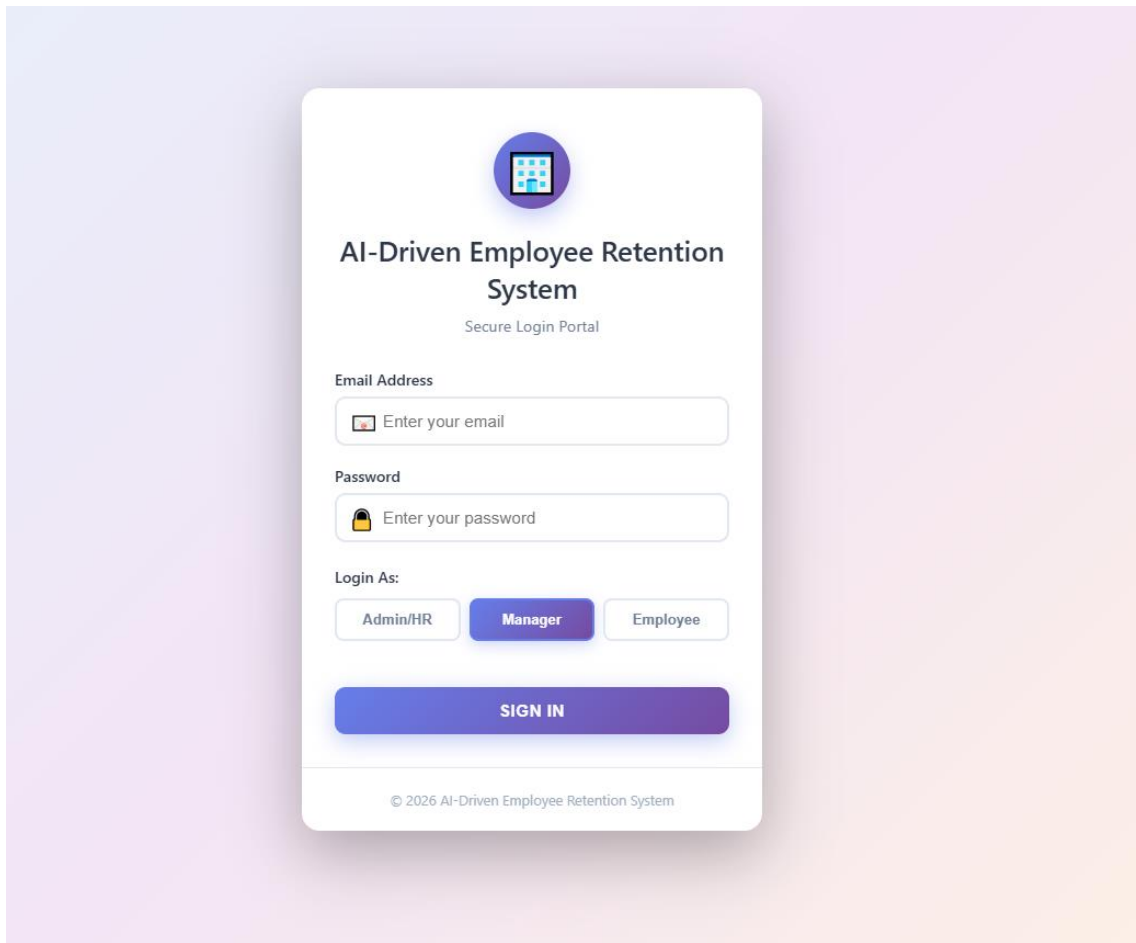


Figure 3.6 Login page

This figure 3.6 shows the login interface used to access the AI-Driven Employee Retention Platform. Users enter their email address and password, then select their role (Admin/HR, Manager, or Employee) before clicking **Sign In**. The role selection ensures the system can route the user to the correct dashboard and display only the functions permitted for that role.

- ADMIN/HR Prototype:

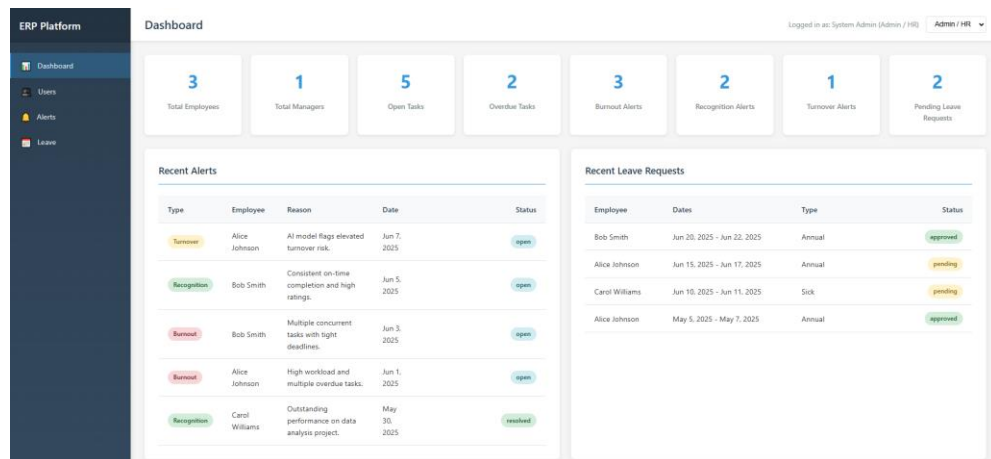


Figure 3.7 Admin/HR Dashboard Page

This figure 3.7 presents the Admin/HR dashboard that summarizes the important counts (employees, tasks, alerts, and pending leave requests) and provides two tables Recent Alerts and Recent Leave Requests so that the Admin/HR could easily monitor the matters and act.

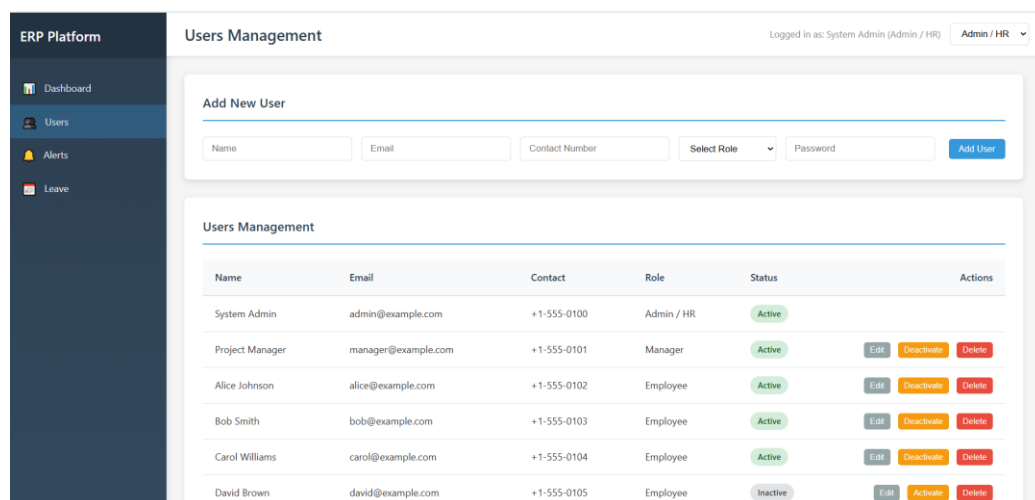


Figure 3.8 User management Page

The figure 3.8 demonstrates the Admin/HR user management module where new users can be added through a simple form, and the existing users can be displayed in a table with their role and status with the options to edit, deactivate/activate, or delete an account.

Type	Employee	Reason	Created Date	Status
Turnover	Alice Johnson	AI model flags elevated turnover risk.	Jun 7, 2025	open

Figure 3.9 Turnover Alert Page

In this figure 3.9, the Turnover Alerts screen appears under the Admin/HR section showing the turnover-related AI alerts in a table with major details like the alert type, employee name, reason, created date, and current status that allow them to monitor easily.

Type	Employee	Reason	Created Date	Status
Burnout	Alice Johnson	High workload and multiple overdue tasks.	Jun 1, 2025	open
Burnout	Carol Williams	Working extended hours consistently.	May 28, 2025	acknowledged
Burnout	Bob Smith	Multiple concurrent tasks with tight deadlines.	Jun 3, 2025	open

Figure 3.10 Need Help Alert Page

This figure 3.10 demonstrates the Need Help Alerts page of the Admin/HR where the Need Help -related alerts are presented in the form of a table with the following important data: type of alert, affected employee, cause, date when it was created, and alert status to facilitate monitoring and subsequent interventions.

ERP Platform	Recognition Alerts				Logged in as: System Admin (Admin / HR)	Admin / HR
	Recognition Alerts					
	Type	Employee	Reason	Created Date	Status	
	Recognition	Bob Smith	Consistent on-time completion and high ratings.	Jun 5, 2025	open	
	Recognition	Carol Williams	Outstanding performance on data analysis project.	May 30, 2025	resolved	

Figure 3.11 Recognition Alert Page

This figure 3.11 displays the Recognition Alerts page under Admin/HR where recognition-related alerts are displayed with the name of the employee, reason, date of creation, and status to assist in keeping track of high-performing employees and assist in taking timely appreciation measures.

ERP Platform	Leave Management				Logged in as: System Admin (Admin / HR)	Admin / HR
	Leave Requests Management					
	Employee	Dates	Type	Reason	Status	Actions
	Alice Johnson	Jun 15, 2025 - Jun 17, 2025	Annual	Family matters	pending	Approve Reject
	Bob Smith	Jun 20, 2025 - Jun 22, 2025	Annual	Personal vacation	approved	
	Carol Williams	Jun 10, 2025 - Jun 11, 2025	Sick	Medical appointment	pending	Approve Reject
	Alice Johnson	May 5, 2025 - May 7, 2025	Annual	Wedding attendance	approved	

Figure 3.12 leave page

This figure 3.12 presents the Admin/HR Leave Management page, which lists employee leave requests with dates, leave type, reason, and current status, and allows Admin/HR to approve or reject pending requests.

- MANAGER Prototype

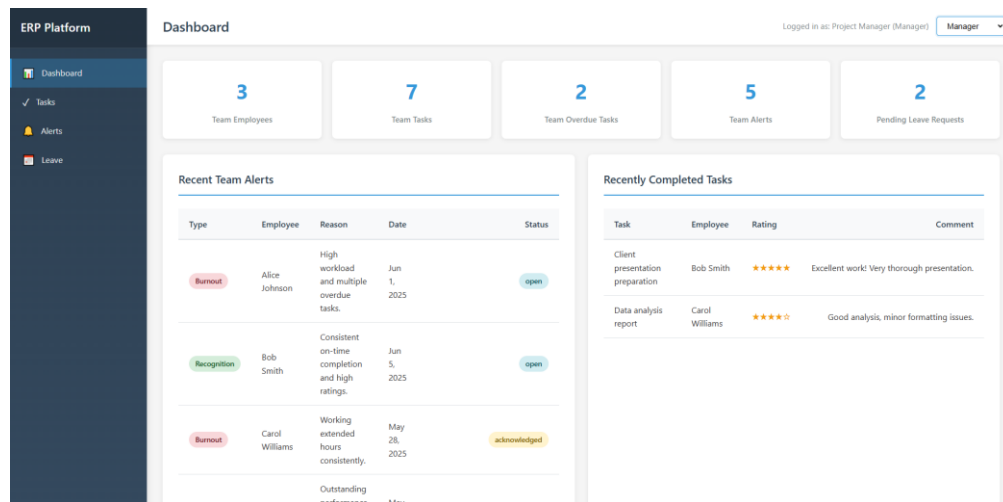


Figure 3.13 Manager Dashboard Page

This is a figure 3.13 of Manager Dashboard, which gives a brief overview of team activity (number of employees, number of tasks, number of tasks due, and alerts, and the number of pending leave requests). It also shows the latest team alerts (Need Help and recognition) and the list of the tasks that were already done with the rating and comments of the manager to monitor the performance.

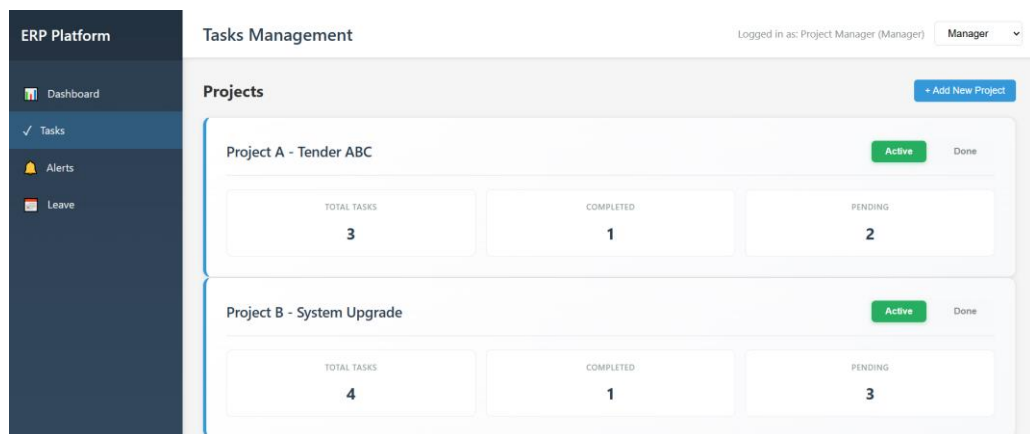


Figure 3.14 Tasks page

This is a figure 3.14 that represents the Tasks Management page of the Manager with projects presented in the form of cards with their brief statistics (total tasks, completed tasks, pending tasks). The manager has an option of adding a new project and switching on and off each project to run or completed.

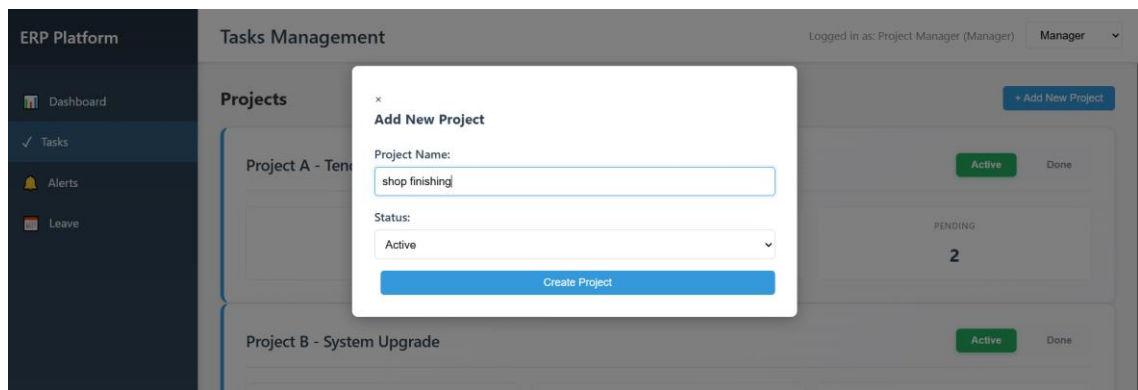


Figure 3.15 Add new project pop up

The figure 3.15 below depicts the pop-up format that the Manager uses to generate a new project. To add the project to the system, the manager inserts the project name, chooses the project status (Active/Done) and sends the form.

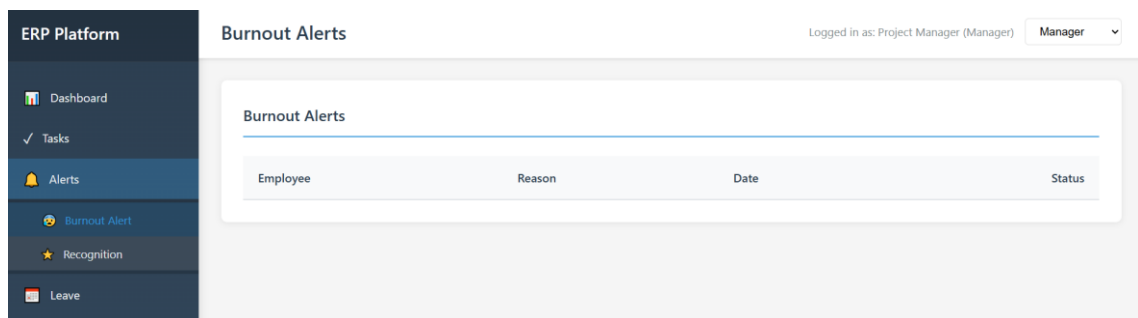


Figure 3.16 Need Help alert page

This figure 3.16 represents the Manager Need Help Alerts page, on which Need Help -related alerts are displayed in a table with the necessary information to track the wellbeing of the team including the employee name, the reason of the alert, the date and status.

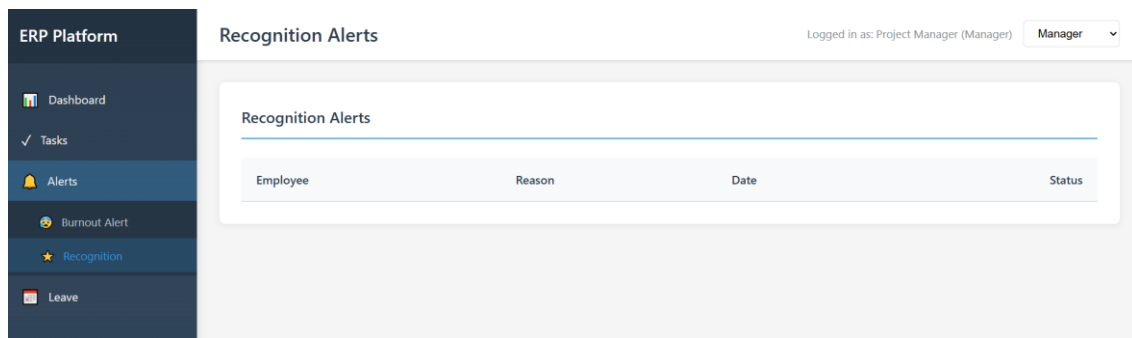


Figure 3.17 Recognition page

This figure 3.17 presents the Manager’s Recognition Alerts page, which lists recognition-related alerts in a table (employee, reason, date, and status) so the manager can identify high-performing employees and take timely appreciation actions.

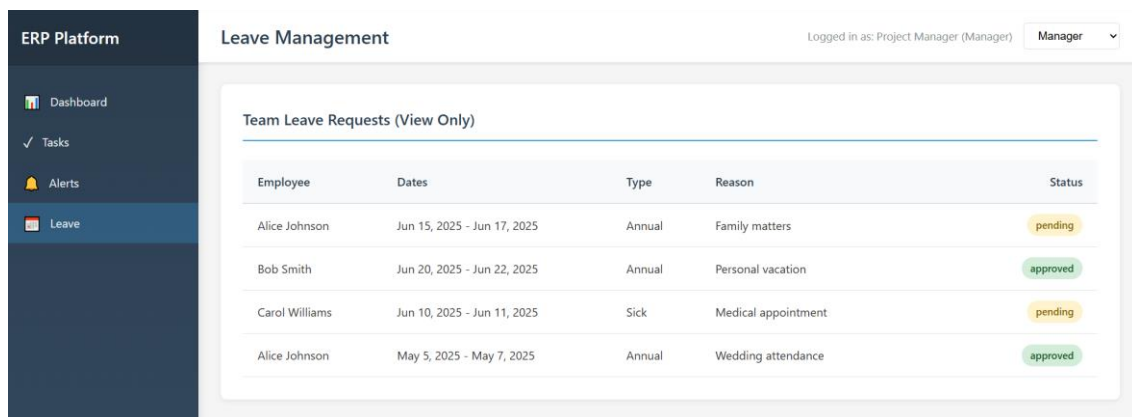


Figure 3.18 Leave page

This figure 3.18 displays the Leave Management page of the Manager in view only mode where the Manager is able to view the team leave requests in a table (employee, dates, leave type, reason and status) to remain updated on the team availability but not approve or reject the requests.

- EMPLOYEE Prototype

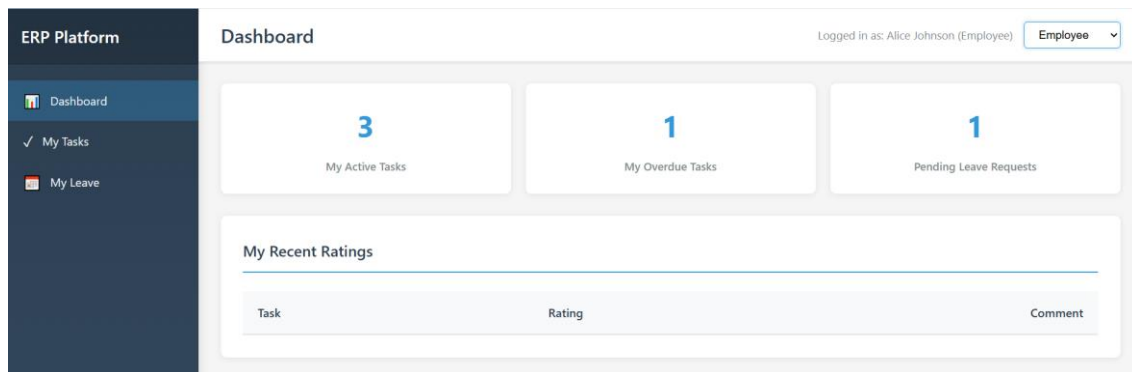


Figure 3.19 Employee dashboard

This figure 3.19 depicts the Employee Dashboard, which gives a basic summary of the workload and position of the employee. It shows summary cards of active tasks, overdue tasks, pending leave requests, and has a Recent Ratings section where the employees will be able to see the most recent task rating and comments of the manager who rated their work.

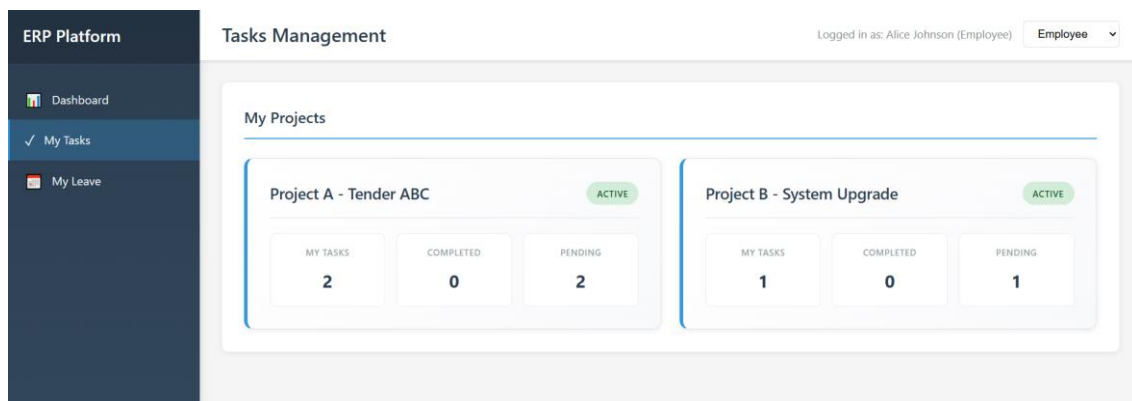


Figure 3.20 My tasks available projects page

This figure 3.20 shows the Employee “My Tasks” page where tasks are organized by project. The employee can view their assigned projects as cards, along with quick statistics for each project (e.g., my tasks, completed, and pending). This layout helps the employee select a specific project and then focus only on the tasks related to that project.

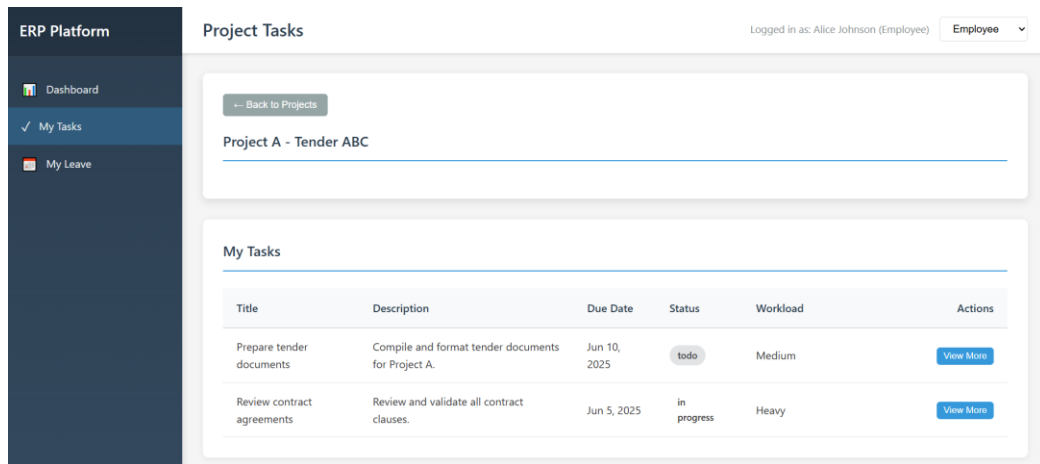


Figure 3.21 My tasks inside project page

This figure 3.21 displays the list of the Employees tasks once they have been chosen in a certain project. The page will show the project name and a table of the tasks that the employee is assigned to work on in that project, the title of the task, the description of the task, the due date, the status, the amount of work, and the Action button. View more option will enable the employee to view the whole task configuration before updating the task progress.

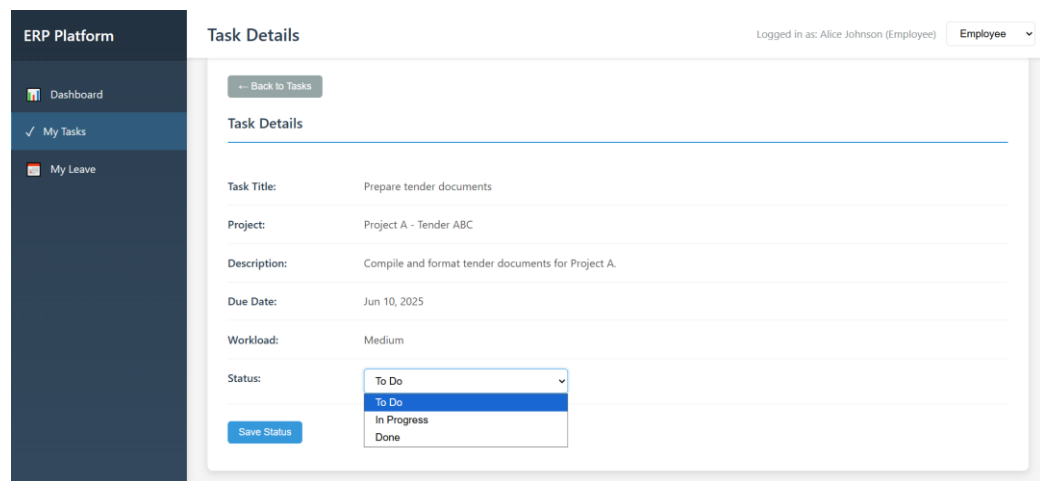


Figure 3.22 My tasks change status page

This figure 3.22 shows the Task Details page for an employee after selecting “View More” from the task list. It displays the full task information such as task title, project, description, due date, and workload. The employee can update the task progress using the status dropdown (To Do / In Progress / Done) and click Save Status to submit the update, ensuring the system records the latest task completion status.

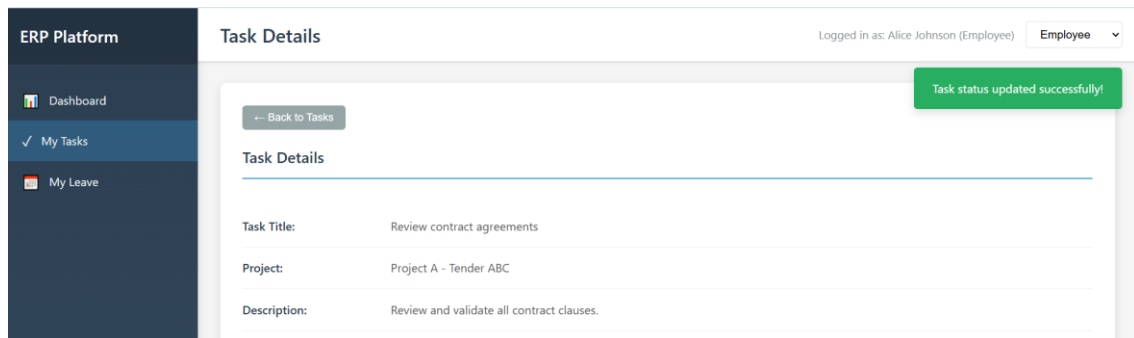


Figure 3.23 My tasks change status successful message

This figure 3.23 shows the confirmation message displayed after an employee updates a task status and clicks Save Status. The green success alert (“Task status updated successfully!”) confirms that the system has saved the new task status, helping the employee know the update was completed correctly.

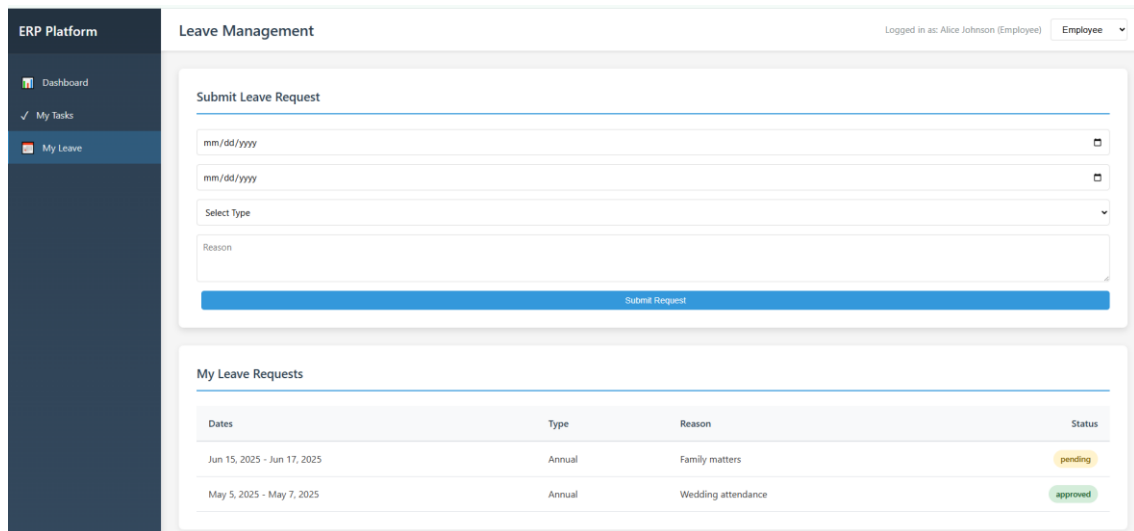


Figure 3.24 My leave page

This figure shows the Employee Leave Management page, where employees can submit a new leave request by entering the leave dates, selecting the leave type, and providing a reason. The same page also displays a table of the employee’s previous leave requests with their current status (e.g., pending or approved), allowing the employee to track updates from Admin/HR.

3.7 Testing/Validation Plan

Table 3.13 presents the key tasks that were performed in the testing phase of the AI-Driven Employee Retention Platform. In the User Acceptance Testing (UAT) phase, the system will be tested by the selected users depending on their roles (Admin/HR, Manager, and Employee) to ensure that all the important functionalities are operating fine and according to the actual workflow of the SME case study [12]. The tasks that the testers will do include creating projects, task assignment, meeting task status, submitting a leave request, approving/rejecting leave request, and seeing the alerts generated by AI. Users will then fill a UAT feedback form after using the system to document their experience with the system, reporting any problems, and recommend improvements. Problems will be corrected and made easier to use with the help of this feedback before the finalizing of the system.

- Testing Strategy / Approach

The AI-Driven Employee Retention Platform testing strategy is a combination of functional and role-based testing, integration testing, and User Acceptance Testing (UAT). The functionality testing is to make sure that every module is functioning properly (users, projects, tasks, ratings, alerts, and leave requests). Role-based testing is used to make sure that the user roles have access to the features corresponding to them (e.g., Admin/HR has the right to approve leave and see all types of alerts, whereas the Manager cannot see turnover alerts). Integration testing: Ensuring that the end-to-end workflow is running smoothly between modules e.g. creating a project, assigning tasks, updating task status, submitting ratings and generating alerts. Lastly, UAT ensures that the system is aligned with the actual SME workflow and can be used by non-technical users.

- Test Data Preparation

A limited amount of realistic sample data will be prepared to be able to run the tests consistently to fit the SME case study. This involves the creation of three user roles (Admin/HR, Manager, and multiple Employees), two projects (one active and one completed), and tasks having various statuses (To Do, In Progress, Done, Overdue) and various workload (Light, Medium, Heavy). Sample leave requests will be inserted with various results (Pending, Approved, Rejected) to verify the handling of leaves. Moreover, sample AI alerts will be added to reflect all the types of system alerts (Need Help, Recognition, and Turnover) to be able to view and filter each role.

- Evaluation Criteria

Pass/fail tests of the test cases as well as the feedback obtained during UAT will be used to validate the system. A successful test case can be defined as one where the actual output matches the expected one (e.g., dashboard redirection is correct, task updates are correct, filtering results are correct, etc.). The UAT participants will give brief comments regarding usability, navigability, and usefulness of alerts and leave control. When the majority of test cases are successfully passed and the UAT feedback shows that the platform can be used to support the desired SME workflow with no serious problems, then the system will be deemed to be ready to be deployed finally.

Table 3.13 (UAT) Test Cases

Topic	Test Case	Input	Process	Expected Result	Success / Fail
Login	User logs into system	Email, password, selected role	User enters credentials → System verifies → Redirects to correct dashboard	User is logged in and redirected based on role	
User Management (Admin/HR)	Add new user	Name, email, role, password	Admin/HR fills form → clicks “Add User”	New user is saved and appears in users table	
User Management (Admin/HR)	Edit user details	Updated name / role / status	Admin/HR clicks “Edit” → updates → saves	User record updates successfully	
User Management (Admin/HR)	Delete user	Selected user	Admin/HR clicks “Delete” → confirms	User removed from the system list	
Project Management (Manager)	Create new project	Project name, status (Active/Done)	Manager clicks “Add Project” → submits form	Project is created and shown in project cards list	
Project Management (Manager)	Toggle project status	Active ↔ Done	Manager changes project status toggle	Project status updates and visibility behaves correctly	
Task Management (Manager)	Create and assign task	Title, description, employee, due date, workload	Manager selects project → clicks “Create Task” → submits	Task is saved under the project and appears in team task table	
Task Management (Manager)	View/filter team tasks	Filter by employee/status	Manager uses filters in team tasks table	Task list updates correctly based on filter	
Task Status (Employee)	View assigned projects	None	Employee opens “My Tasks”	Employee sees only assigned projects	

Task Status (Employee)	Update task status	Selected task + new status (To Do/In Progress/Done)	Employee opens task details → updates status → submits	Task status updates successfully and reflects in task list	
Task Rating (Manager)	Rate completed task	Rating (1–5), comment	Manager opens rating table → rates task → submits	Rating and comment are saved and visible to employee	
Alerts (Admin/HR)	View all alert types	Alert type filter (Turnover / Need Help /Recognition)	Admin/HR opens alerts page → filters	System displays correct alert list by selected type	
Alerts (Manager)	View team alerts (limited types)	Alert type filter (Need Help /Recognition)	Manager opens alerts page	Manager sees only Need Help & Recognition alerts (no turnover)	
Leave Requests (Employee)	Submit leave request	Start date, end date, leave type, reason	Employee fills form → submits	Leave request saved with “Pending” status	
Leave Requests (Admin/HR)	Approve/reject leave	Decision (Approve/Reject)	Admin/HR opens leave list → selects request → approves/rejects	Leave request status updates correctly	
Leave Requests (Manager)	View leave requests (read- only)	None	Manager opens leave page	Manager can view team leave requests but cannot approve/reject	

3.8 Potential Use of Proposed Solution

The intended outcome of the project will be to assist SMEs in alleviating Need Help and turnover through providing managers with early warnings and clear information rather than responding to it once the staff has become stressed or opted to leave [5]. Within a real tender-company setup, the platform will be utilized in the day-to-day operations of managing projects, tasks, task ratings, and leave requests within a single system. The system follows the trends in KPI (e.g., overdue work, high workload, rating trends, and frequency of leaves) as employees work and managers rate their performance and issues alerts related to Need Help, recognition, and turnover risk based on AI. This aids the Admin/HR and Manager to act sooner like redistributing, altering the deadline, offering assistance or rewarding good performers before the situation turns out to be serious. The key advantages are increased team stability, reduced turnover, and reduced time and expenditure in recruiting new employees and increased equitability since decisions are made based on facts and not subjective opinions [1].

To be commercialized the platform may be packaged as a light-weight web solution to SMEs in time-based industries (tendering, consulting, agencies, and small project teams). The use of advanced HR analytics tools is not common by many SMEs; therefore, a basic dashboard that would integrate task tracking with explainable AI alerts could be a useful and scalable subscription-based product [4].

CHAPTER 4

RESULTS AND DISCUSSION

4.1 INTRODUCTION

The chapter includes the implementation, testing, and results of the AI-Driven Employee Retention Platform. The system functions are described in terms of the methodology, requirements, and design of the last chapter. The platform combines task management, leave management, performance rating, and AI-based alerts to assist in employee retention in SMEs. Moreover, the rule-based Need Help and recognition are tested along with the Logistic Regression turnover prediction model to ensure the functionality, reliability, and the overall effectiveness of the system.

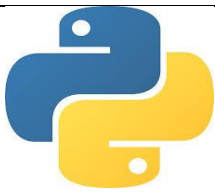
4.2 Implementation & Development

This section outlines the key steps in the creation of the AI-Driven Employee Retention Platform. It starts with the development environment setup, such as the frontend, backend, database, and Python-based machine learning setup. The core system, including user authentication, user and project management, task assignment, leave management, performance rating, and AI alert generation, is then rolled out in stages. In the development process, modules are tested to verify their functionality, usability and integration to produce a working prototype that can be evaluated and deployed.

4.2.1 Development Process & Tools

The AI-Driven Employee Retention Platform was developed with the help of MERN stack, which provides high-quality and efficient functionality. The instruments and technologies applied to the development process are summarized in Table 4.1 below.

Table 4.1 Development Process & Tools

NO	Tool	Description
1		An interactive and responsive user interface library used to build the frontend pages of the AI-Driven Employee Retention Platform, including dashboards, task views, leave management, and alert displays.
2		A runtime environment used to execute JavaScript on the server side and handle backend logic, asynchronous operations, and communication between the frontend, database, and AI components.
3		A lightweight backend framework used to build RESTful APIs, manage routes, process client requests, and connect the system with the database and machine learning module.
4		A relational database management system used to store and manage structured system data such as users, projects, tasks, ratings, alerts, and leave requests reliably and efficiently.
5		A programming language used to implement the machine learning component of the platform, particularly for preparing synthetic data, training the prediction model, and generating turnover-risk outputs.

4.3 System Components & Functions

4.3.1 Login

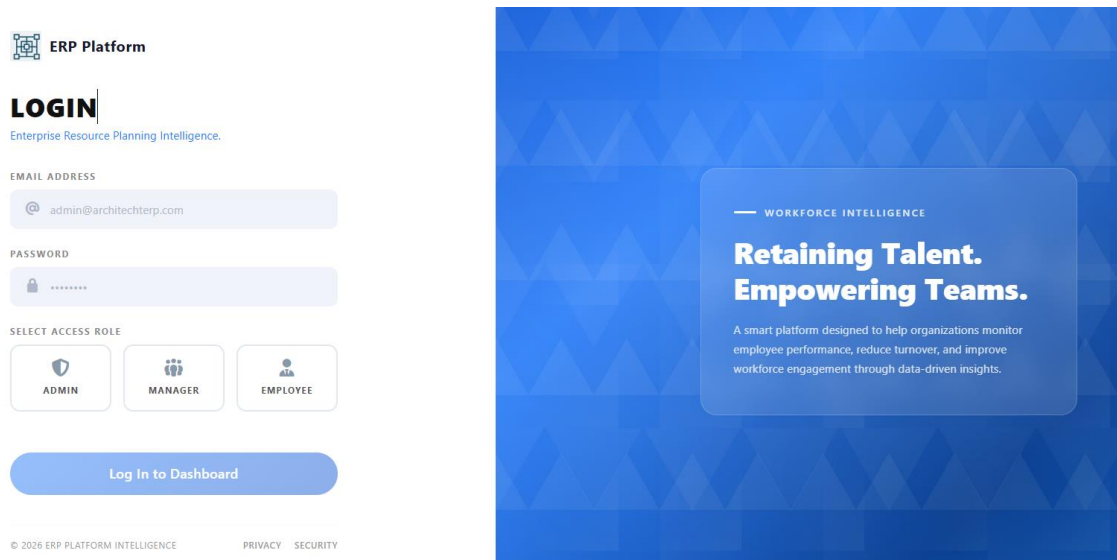


Figure 4.1 Login Page

The Login page will be the primary access point to the AI-Driven Employee Retention Platform to all user types, i.e., Admin, Manager, and Employee. Before logging in, users must provide their email address and password and choose the right access role. The page also displays the platform branding and a brief introduction that emphasizes the purpose of the system in helping to monitor the performance of employees and reduce turnover. After verifying the credentials, the user is redirected to the respective dashboard according to role-based access control, so that only the respective roles can access the functions and data they are allowed to.

ADMIN:

4.3.2 Admin Dashboard

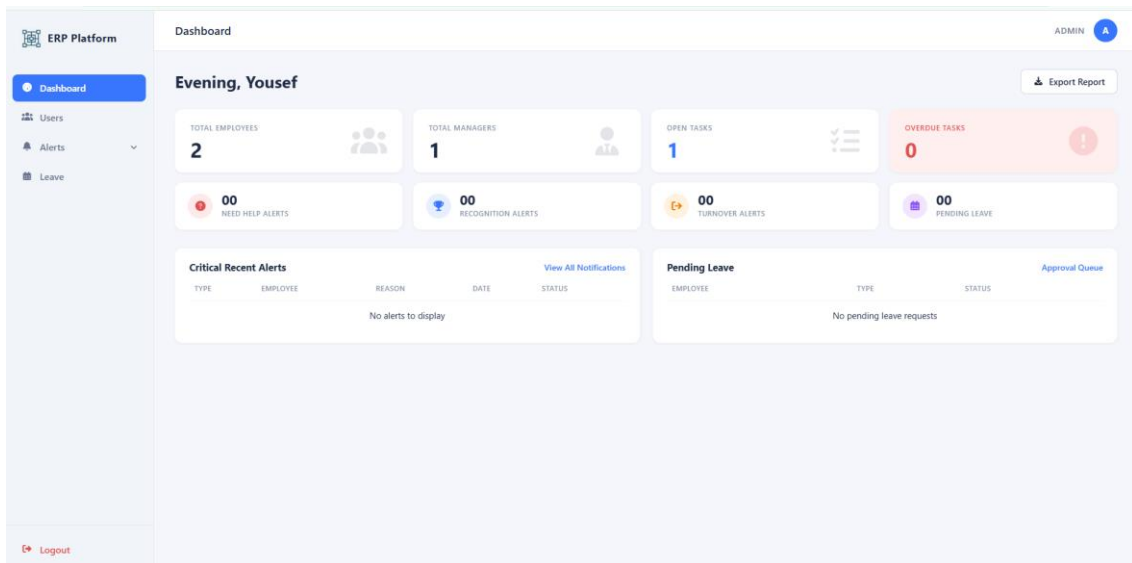


Figure 4.2 Admin Dashboard Page

The Admin Dashboard page is a centralized view of the most significant system data and AI-related indicators within the AI-Driven Employee Retention Platform. It shows summary cards of total employees, total managers, open tasks, overdue tasks, need help alert, recognition alert, turnover alert and pending leave request, enabling the Admin to view the current state of the organization at a glance. Two major tables are also provided on the page, i.e., Critical Recent Alerts and Pending Leave, which facilitate the rapid overview of the employee-related problems and leave actions.

4.3.3 User Management page

The screenshot displays the 'Users' management interface. The sidebar on the left contains links for 'Dashboard', 'Users' (highlighted), 'Alerts', and 'Leave'. The main content area features a top navigation bar with 'Users' and an 'ADMIN' user profile. Below this is an 'Add New User' form with the following fields: 'FULL NAME' (placeholder: e.g. Julian Casablancas), 'EMAIL ADDRESS' (placeholder: yousef@admin.com), 'CONTACT NUMBER' (placeholder: +1 (555) 000-0000), 'ROLE PROFILE' (dropdown menu with 'Employee' selected), and 'INITIAL PASSWORD' (masked with dots). An 'Add User' button is located to the right of the form. Below the form is an 'Active Directory' table with the subtitle 'Manage permissions and account lifecycles'. The table has columns for 'IDENTITY', 'CONTACT DETAIL', 'ACCESS ROLE', 'ACCOUNT STATUS', and 'ACTIONS'. It lists three users: 'ali' (Employee, Active), 'aziz alawadi' (Employee, Active), and 'mohammed salah' (Manager, Active). Each user row has 'EDIT', 'DEACTIVATE', and 'DELETE' action buttons. At the bottom of the table, it says 'Showing 3 of 3 system users' with 'Previous' and 'Next' pagination buttons. A 'Logout' button is located at the bottom left of the sidebar.

IDENTITY	CONTACT DETAIL	ACCESS ROLE	ACCOUNT STATUS	ACTIONS
ali ali@gmail.com	01122334455	EMPLOYEE	Active	EDIT DEACTIVATE DELETE
aziz alawadi aziz@gmail.com	01198765432	EMPLOYEE	Active	EDIT DEACTIVATE DELETE
mohammed salah msd@gmail.com	0112345678	MANAGER	Active	EDIT DEACTIVATE DELETE

Figure 4.3 Users Management Page

The User Management page allows the Admin to control employee and manager accounts in the AI-Driven Employee Retention Platform. The Add New User form, which is at the top of the page, enables the Admin to add user information including full name, email address, contact number, role profile, and initial password to create a new account. The list of the existing system users along with their identity, contact details, access role, and account status are shown in the Active Directory table below the form. The page also offers action options like Edit, Deactivate, and Delete, which enable the Admin to keep user information and manage account availability effectively.

4.3.4 Turnover/Need Help Alerts Page

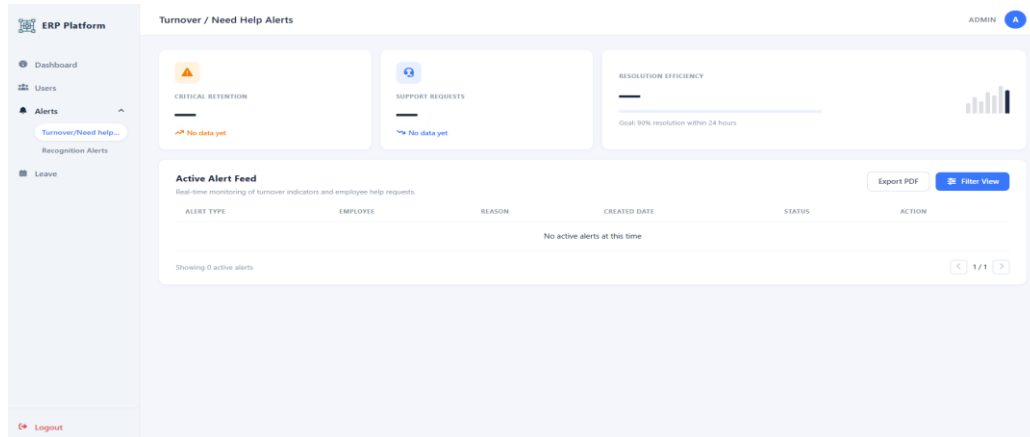


Figure 4.4 Turnover/Need Help Alerts Page

The Turnover / Need Help Alerts page enables the Admin to view employee retention-related alerts that the system has raised. Summary cards at the top of the page display the main information like Critical Retention, Support Requests, and Resolution Efficiency that give a brief overview of the current alert situation. The Active Alert Feed is the primary part of the page, and it shows the alert records in a form of a structured table with the alert type, employee, reason, created date, status and action. Moreover, the page also offers Export PDF and Filter View features to facilitate reporting and effective review of alert data to make decisions and intervene.

4.3.5 Recognition Alerts Page

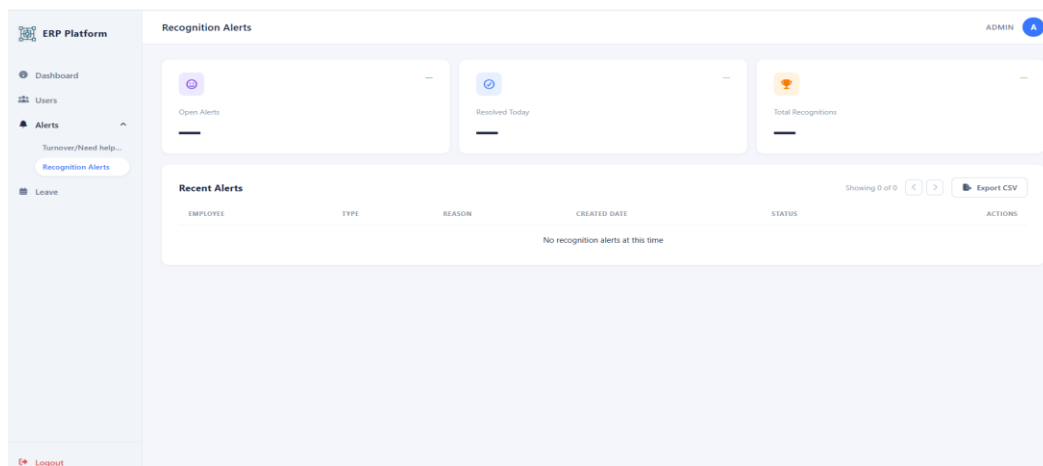


Figure 4.5 Recognition Alerts Page

Turnover / Need Help Alerts page enables the Admin to view alerts related to employee retention created by the system. Summary cards that display the most important information, including Critical Retention, Support Requests, and Resolution Efficiency, are placed at the top of the page and allow seeing the current alert situation in a brief overview. The primary part of the page is the Active Alert Feed, which shows the records of alerts in a tabular format with the alert type, employee, reason, created date, status, and action. Moreover, the page also offers Export PDF and Filter View capabilities to facilitate reporting and effective review of alert data to make decisions and intervene.

4.3.6 Leave Management Page

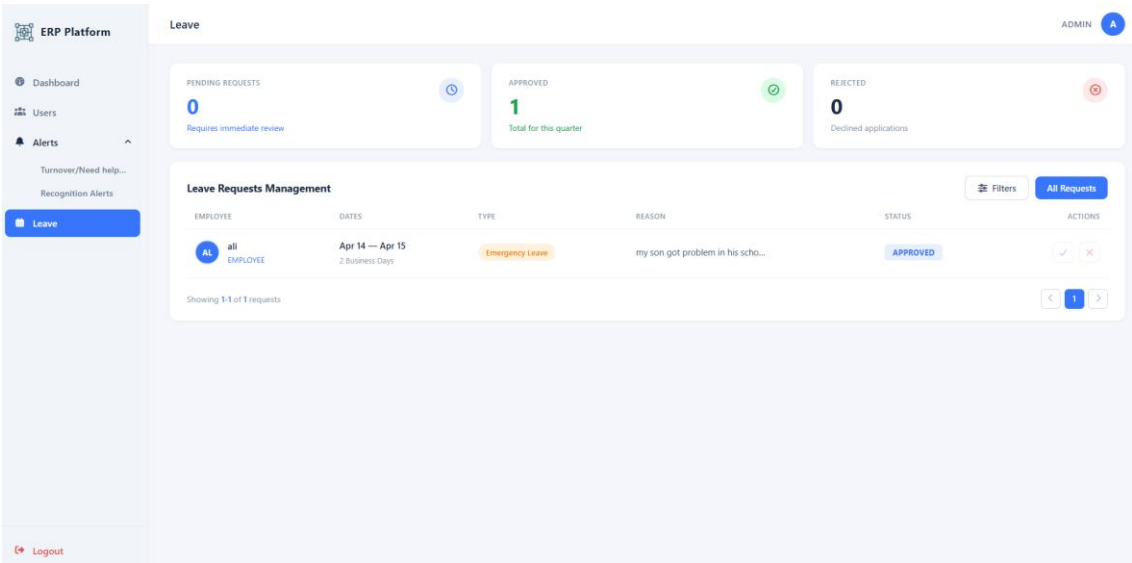


Figure 4.6 Leave Management Page

The Leave Management page enables the Admin to track and manage employee leave requests in the AI-Driven Employee Retention Platform. Summary cards at the top of the page show the number of Pending Requests, Approved and Rejected leave applications, which give a brief overview of the leave activity. The central part of the page is the Leave Requests Management table that displays the record of employee leaves along with the leave dates, the type of leave, the reason, the status, and the action buttons. There are also Filters and All Requests functions in the page to facilitate easier review and management of leave applications.

Manager:

4.3.7 Dashboard Page

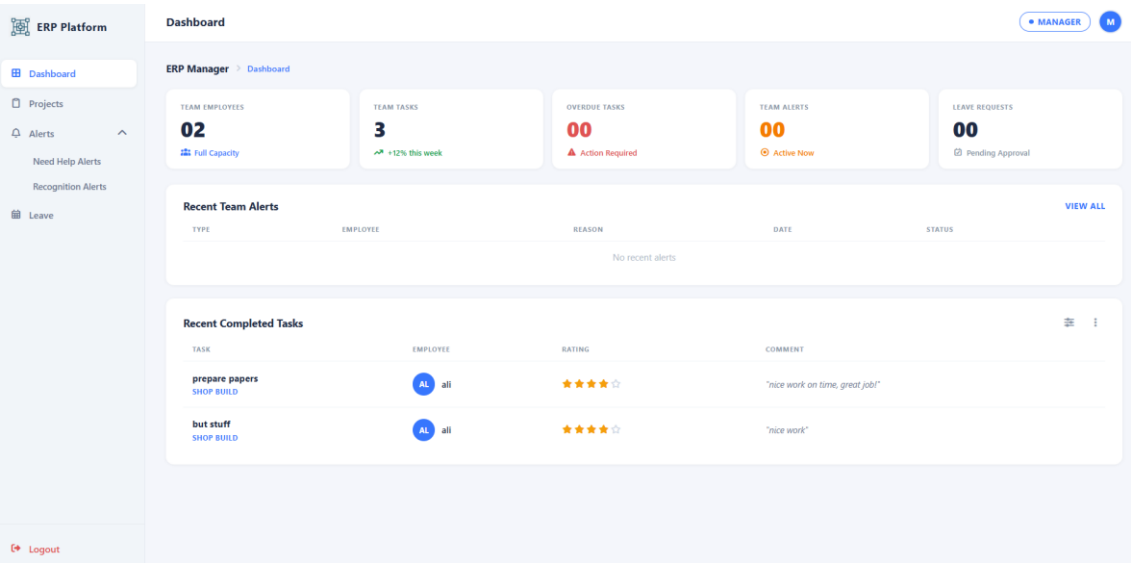


Figure 4.7 Dashboard Page

Manager Dashboard page is a summary of the team-related activities and performance in the AI-Driven Employee Retention Platform. Summary cards at the top of the page show important information like Team Employees, Team Tasks, Overdue Tasks, Team Alerts, and Leave Requests, which enable the Manager to easily check the status of the team at hand. There is also a Recent Team Alerts table to review the most recent Need Help and recognition-related notifications and a Recent Completed Tasks table that displays the completed team tasks along with the assigned employee, rating, and manager comments. This dashboard helps the Manager to monitor the productivity of the team, issues, and review the completed ones.

4.3.8 Project Pages

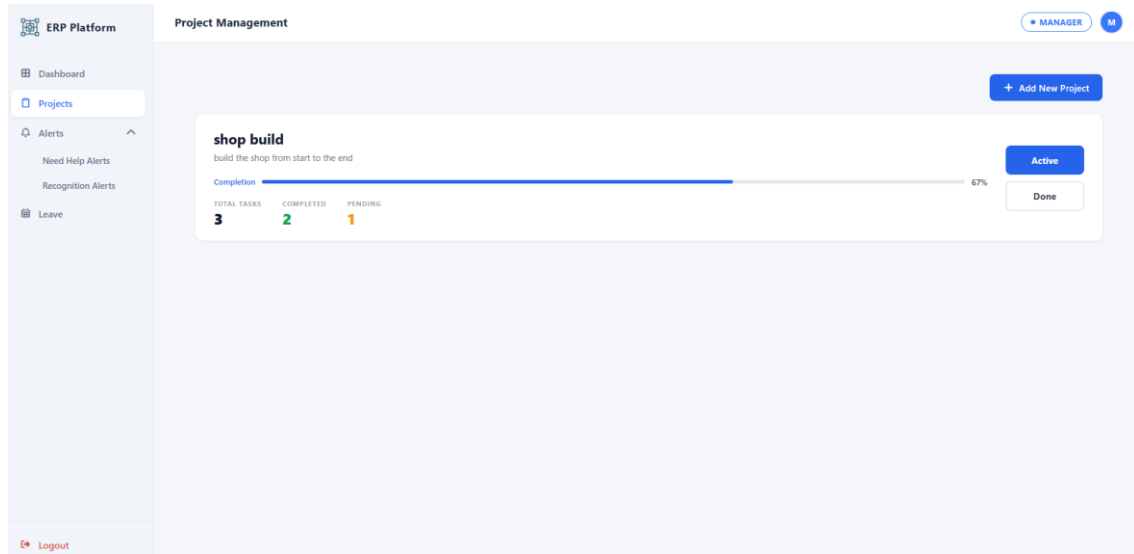


Figure 4.8 project page

The Project Management page enables the Manager to track all the available projects and manage their status in the AI-Driven Employee Retention Platform. The projects are presented in a card format with project title, brief description, progress of completion, total number of tasks, tasks completed and those pending. The page also has an Add New Project button where the Manager can create new projects and the Active and Done controls that enable the Manager to control the visibility and progress of the projects effectively.

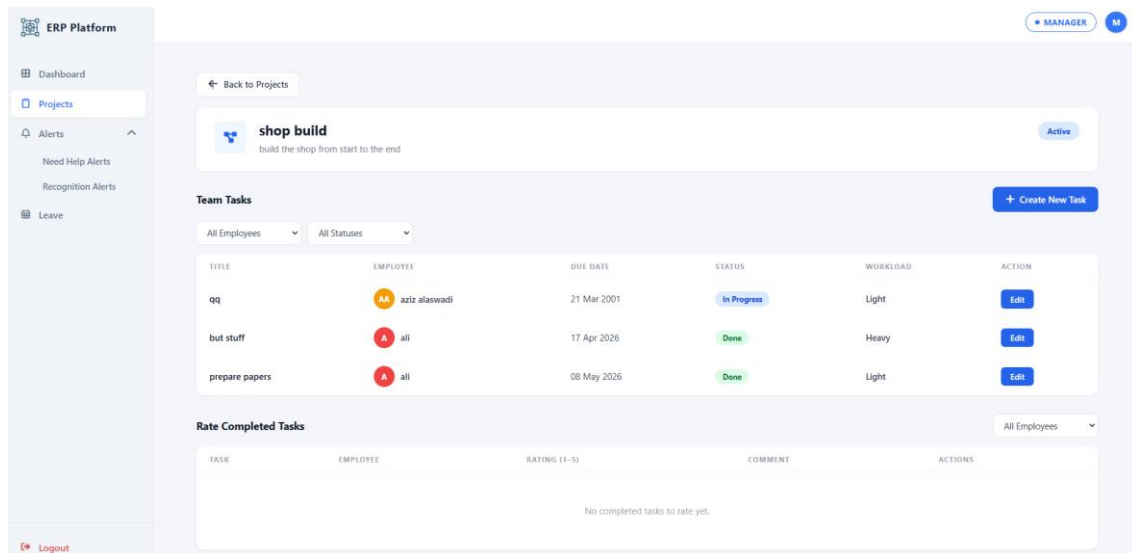


Figure 4.9 Tasks Page

The Team Tasks page enables the Manager to see and control tasks within a chosen project. The chosen project information is presented at the top of the page along with its current status. The Team Tasks section displays all the tasks assigned in a table format with the title of the task, employee, due date, status, workload, and action button. The page also has the filtering options like All Employees and All Statuses to help in easily monitoring tasks. Moreover, the Rate Completed Tasks section allows the Manager to view and rate completed tasks of employees, which assists in supporting performance appraisal and feedback in the section.

4.3.9 Need Help Alerts Page

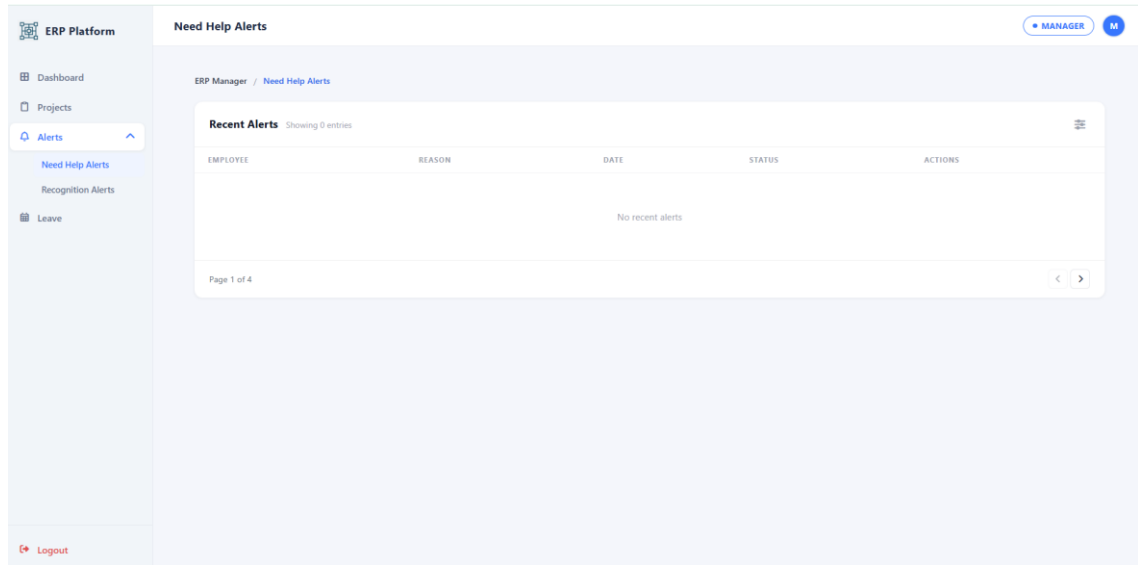


Figure 4.10 Need Help Alerts Page

The Need Help Alerts page enables the Manager to track support-related alerts of team members in the AI-Driven Employee Retention Platform. The page displays a Recent Alerts table which sorts the records of the alerts by employee, reason, date, status and actions available so that the Manager can easily view the employees who might need attention or intervention. This page will assist the Manager to monitor the well-being of the team and address the needs of employees in a timely manner.

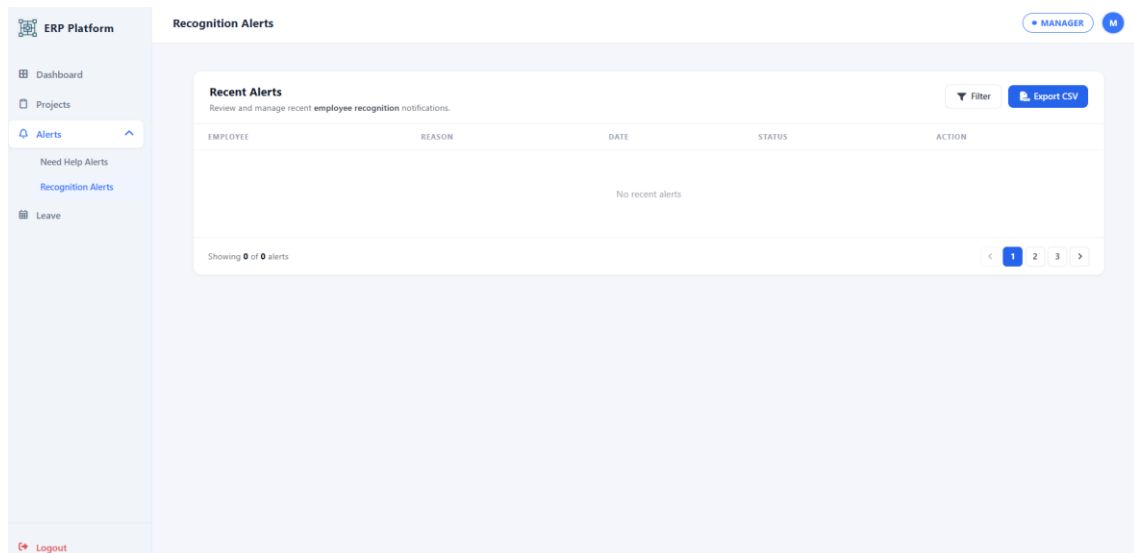


Figure 4.11 Recognition Page

Recognition Alerts page enables the Manager to view employee appreciation and performance-related recognition alerts that are created by the system. The primary part of the page is a Recent Alerts table that shows the name of the employee, the reason, date, status, and what actions can be taken with each recognition record. Moreover, the page offers Filter and Export CSV options to assist the Manager to review, organize, and report information related to recognition more effectively.

4.3.10 Team Leave Request Page

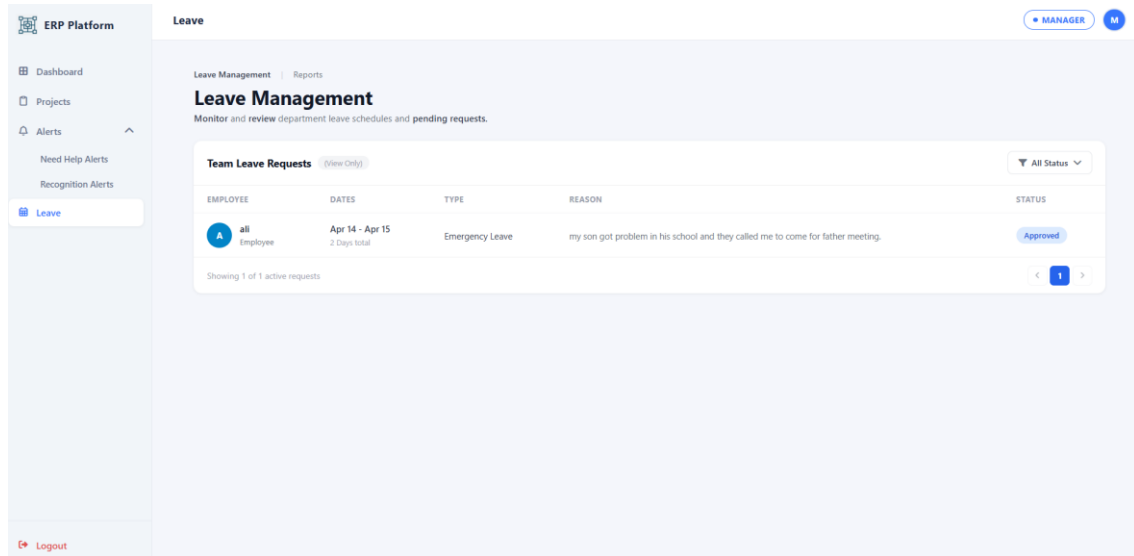


Figure 4.12 Team Leave Request page

The Team Leave Requests page enables the Manager to view leave requests made by the team members in a read-only format. The page includes a table of Team Leave Requests with the following information: employee name, leave dates, type of leave, reason, and approval status, which allows the Manager to keep track of the availability and absence schedule of the team. Moreover, the page also has a status filter, which helps to review leave records more easily, depending on their approval status.

Employee:

4.3.11 Employee Dashboard Page

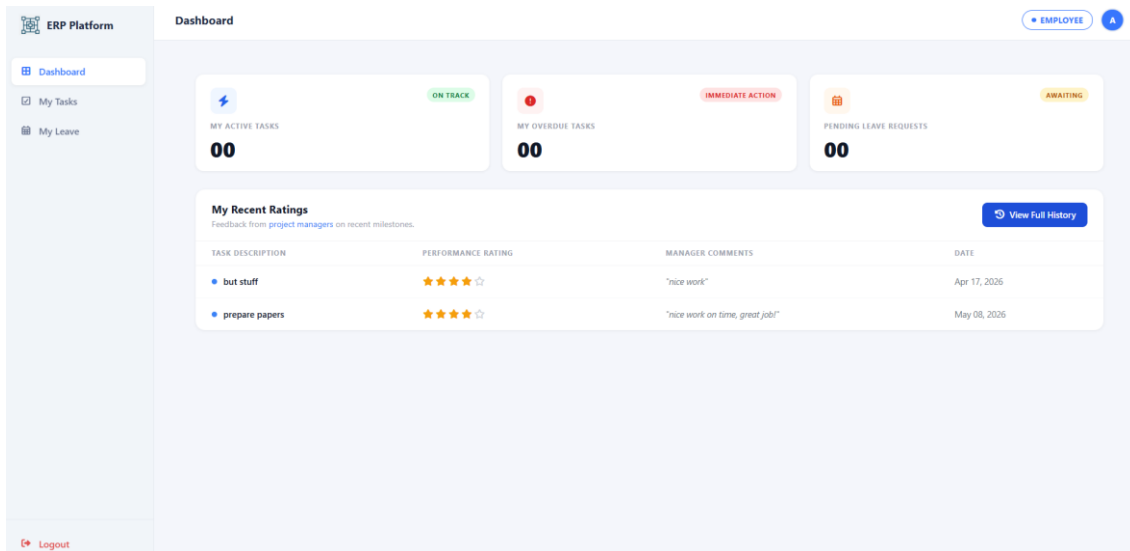


Figure 4.13 Employee Dashboard Page

The Employee Dashboard page gives a personalized view of the current work and leave position of the employee in the AI-Driven Employee Retention Platform. Summary cards are shown at the top of the page, containing the most important information, including My Active Tasks, My Overdue Tasks, and Pending Leave Requests, enabling the employee to quickly comprehend what he/she has to do and what is pending. My Recent Ratings is located in the lower part of the page and it displays recently rated tasks along with performance ratings, manager comments and dates. This dashboard assists employees to be aware of their performance feedback and the status of their tasks.

4.3.12 My Tasks Page

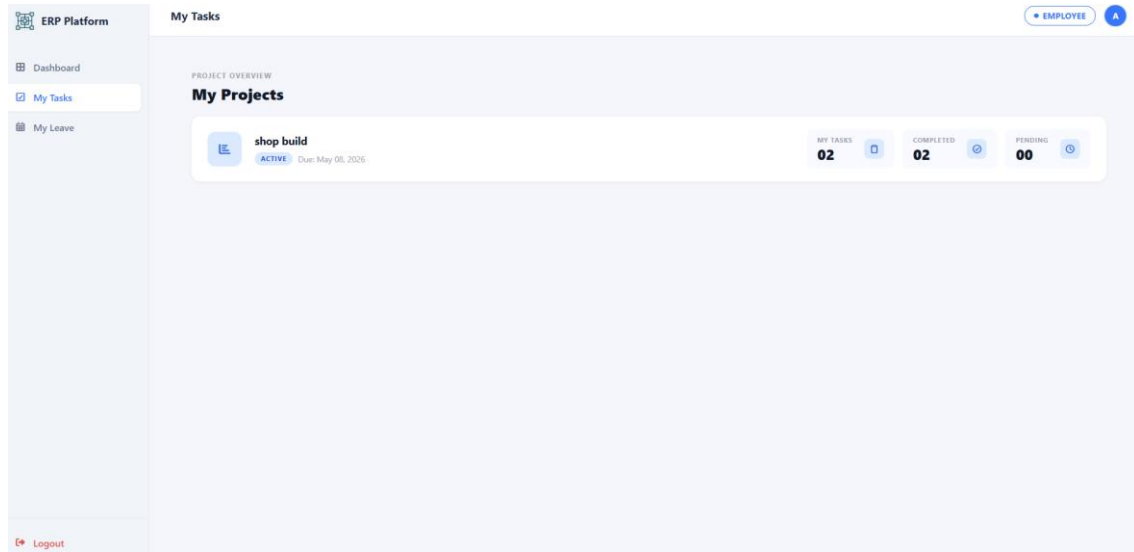


Figure 4.14 My Tasks Page

My Tasks page enables the employee to see the projects where they have been assigned the tasks. The projects are presented in a card format with the project name, status, due date and quick task statistics including My Tasks, Completed and Pending. This page assists the employee to recognize the active assigned projects and choose the project to review and manage their task progress more conveniently.

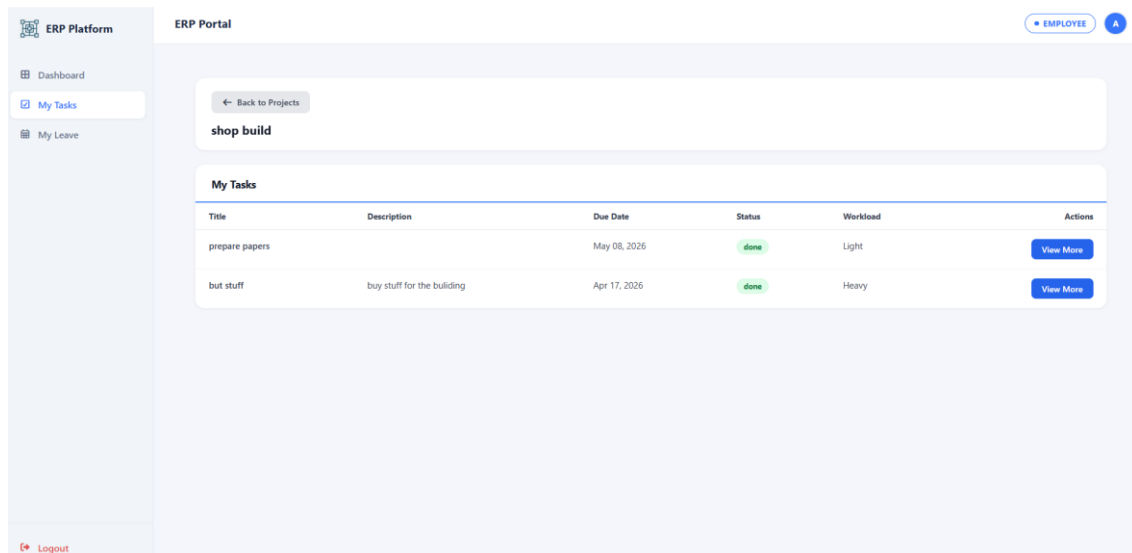


Figure 4.15 Inside My Tasks page

My Leave page enables the employee to make new leave requests and see the history of their past leave requests. The leave request form has the employee filling in the From Date, To Date, Type of leave and the Reason why the leave is required on the top of the page and then submitting the request to the system. The bottom part, My Leave Requests, shows the history of leave requests of the employee in a table with the date of request, the dates of leave, the type of leave, the reason, and the status of the leave request. This page allows employees to handle leave requests and monitor the results of approval easily.

4.3.13 My Leave Page

The screenshot displays the 'My Leave' page within an 'ERP Platform'. The left sidebar contains navigation links: 'Dashboard', 'My Tasks', and 'My Leave' (which is highlighted). The main content area is titled 'My Leave' and includes a user profile 'EMPLOYEE' in the top right corner. The primary section is a form for requesting leave, featuring 'FROM DATE' and 'TO DATE' input fields with date pickers, a 'LEAVE TYPE' dropdown menu set to 'Annual Leave', and a 'REASON FOR LEAVE' text area with a placeholder 'Please provide a brief reason for your request...'. A 'Submit Request' button is located at the bottom right of the form. Below the form, a 'HISTORY' section titled 'My Leave Requests' contains a table with the following data:

REQUEST DATE	DATES	TYPE	REASON	STATUS
Apr 13, 2026	Apr 14 – Apr 15 2 Working Days	Emergency Leave	my son got problem in his school and they called me to come for father meeting.	APPROVED

A 'Logout' link is visible in the bottom left corner of the sidebar.

Figure 4.16 My Leave Page

My Tasks Inside Project page enables the employee to see all the tasks which have been assigned to him or her under a chosen project. The employee can view the project name selected at the top of the page and can go back to the project overview page by clicking Back to Projects button. The primary section shows a My Tasks table with the title of tasks, description, due date, status, workload, and an action button. View More feature enables the employee to open a particular task and see its details in greater detail before making any changes to the task progress.

4.4 Testing

- Functional Testing:

Functional testing was conducted to ensure that the key functions of the AI-Driven Employee Retention Platform work as per the requirements. The testing was centred on the fundamental modules that were applied in the system such as authentication, user management, project management, task management, task rating, leave management and viewing of alerts. All functions were tested by giving valid inputs and seeing whether the real system output was the same as the expected one. This testing will assist in ensuring that the platform is capable of supporting the desired workflow of the Admin/HR, Manager, and Employee users in a proper and trustworthy way.

Table 4.2 Functional Testing

No	Module / Function	Test Case	Input / Action	Expected Result
1	Authentication	User login with correct credentials and role	Valid email, password, selected role	User is successfully logged in and redirected to the correct dashboard based on role
2	User Management	Add new user	Name, email, contact number, role, password	New user record is created and displayed in the users list
3	User Management	Edit / deactivate / delete user	Update user details or choose deactivate/delete action	Selected user account is updated, deactivated, or removed successfully
4	Project Management	Create new project	Project name and description, then click Add New Project	New project is created and shown in the project list
5	Project Management	Change project status	Select Active or Done for a project	Project status is updated correctly and reflected in the interface

6	Task Management	Create and assign task	Task title, description, due date, workload, assigned employee	Task is saved under the selected project and appears in the team task list
7	Task Management	Employee views assigned tasks	Employee opens My Tasks and selects a project	Employee sees only the tasks assigned to them under that project
8	Task Management	Update task status	Employee selects task and changes status	Task status is updated successfully and reflected in the task list
9	Task Rating	Manager rates completed task	Rating value and comment for a completed task	Task rating and comment are saved and become visible in the employee dashboard/history
10	Leave Management	Submit leave request	Leave dates, leave type, reason	Leave request is stored successfully with the correct status
11	Leave Management	Approve / reject leave request	Admin/HR selects approve or reject action	Leave request status is updated correctly in the system
12	Alerts	View turnover / need help / recognition alerts	User opens alert page according to role	Relevant alerts are displayed correctly based on the user role and page type

CHAPTER 5

CONCLUSION

5.1 Introduction

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